

When Does a Brand-Influencer Matching AI Backfire? *

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Abstract

We consider a social media platform that offers a matching service to match marketers with influencers through Artificial Intelligence (AI) technology. We find that, even if the implementation cost is negligible, it is not always in a platform’s best interest to adopt such AI technology or to perfect its AI accuracy. The results arise from two countervailing effects on the participation incentives of influencers, which in turn affects the platform’s profit. On the one hand, influencer marketing generates higher sales from a better influencer-marketer match, which benefits both influencers and the platform as they share a commission proportional to sales. On the other hand, if sales is high via an influencer’s recommendation channel, more users may stop following the influencer if they bought a low-quality product. The former “*sales effect*” induces the influencer to participate in a marketing campaign, whereas the latter “*quality concern effect*” deters one from doing so. Furthermore, the proportional size of loyal followers moderates the trade-off between these two effects. We derive conditions under which adopting such AI technology is profitable for a platform. We also extend our baseline model to study when and how a platform should integrate its AI strategy with its communication strategy for product quality assurance.

Keywords: Artificial intelligence, Influencer marketing, Information design

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1 Introduction

Social media usage is up by almost 60% during the pandemic lockdown (Nelson Media Report, 2020). With that, influencer marketing – where marketers collaborate with Internet “influencers” to promote their products – has now become one of the top trends in online advertising. Such influencers have a significant number of followers consuming the content they produce on social media platforms, thus making them an ideal conduit to potential customers. When collaborating with marketing firms, the influencers create specific “content” (such as videos, blogs, etc.) to reach targeted customers, drive engagement, and ultimately convert sales. It is expected that marketing managers will spend \$15 billion on influencer marketing by 2022 (Avery and Israeli, 2020). Meanwhile, social media platforms also benefit from influencer marketing because they now have a new source of revenue from charging service fees for those on-the-platform marketing campaigns. For example, TikTok’s Chinese version Douyin and its major competitor Kuaishou charge a commission of 2-5% from influencers’ on-the-platform campaigns in the form of “technical service fees” (Southern Metropolis Daily, 2020).

Since influencers have widely heterogeneous characteristics, the effectiveness of a marketer’s influencer marketing campaign depends critically on her choice of the “right” influencer. Avery and Israeli (2020, p. 7) point out in a Harvard Business School technical note on influencer marketing that “according to marketers, one of the most frustrating challenges associated with influencer marketing is choosing the right influencers with whom to work.” The difficulty of finding the “right” influencer lies in the fact that there are a large and increasing number of influencers, especially micro-influencers, available on the Internet. For example, by October 2020, there were 500,000 active influencers on Instagram¹, and in 2017, there were 6.4 million Twitter users who had 1K-10K followers, which comprise 2% of the total user base on Twitter². One proposed solution to improving the matching between marketers and influencers is for platforms to offer artificial intelligence (AI) or machine learning algorithms (Sweet et al., 2019; Farseev et al., 2018). Social media platforms care about the match between brands and influencers because they collect a percentage service fee from sales induced by on-the-platform influencer marketing. Having said that, should platforms embrace such AI technology that helps marketers find the “right” influencers? Specifically, when is adopting such AI technology profitable for a platform? When it is, how should a

¹<https://www.omnicoreagency.com/instagram-statistics/>.

²<https://askwonder.com/research/social-influencers-10k-100k-followers-world-us-jzrt515im#>.

platform optimally design its prediction accuracy? How does the implementation of such AI technology influence the pricing and communication strategies of the platform?

To address these questions, we develop a theoretical model that captures four salient features of influencer marketing. First, when participating in on-the-platform marketing campaigns, influencers engage their followers through videos or blogs and offer direct shop-from-content capabilities. Consumers can purchase the featured product without having to leave the platform, making the content “seamlessly shoppable” (Avery and Israeli, 2020). We model this by assuming that influencers’ revenue from on-the-platform marketing campaigns is proportional to their converted sales. Second, most influencers have *some* “loyal” followers who purchase products endorsed by the influencers just to show their support, regardless of how much they like the product per se. According to a 2019 survey about influencer marketing on Kuaishou³, 12% of the users have purchased products endorsed by an influencer simply because they wanted to “support the influencer.”⁴ Avery and Israeli (2020, p. 10) also point out that “influencers must have a sizeable and loyal audience to be attractive to marketers.” We capture this phenomenon in our model by assuming that a proportion of an influencer’s followers always purchase the endorsed product. Third, followers may lose trust from the influencer and react negatively when they receive a low-quality product. However, it is hard for an influencer to ensure that every product her followers purchased via her endorsement channel is high-quality. For instance, Kuaishou’s top influencer Xinba was caught selling fake edible bird’s nest to his followers and is now under investigation (Yi, 2020). To reflect this reality, our model assumes that followers who receive a low-quality product may stop following an influencer, which negatively impacts the influencer’s future value. Fourth, influencers differ in their content positioning: some influencers create content focused on “a cluster of subjects” inside a specific domain, for example, Amanda Julia, a Canadian influencer, manages an Instagram channel that is mainly about horses⁵. Others “transit between different topics and off-topics.” For example, Laura Hunt, a lifestyle influencer, describes her Instagram posts as “pieces of my life displayed in grids⁶.” In our model, we refer to the first type of influencers as “niche” influencers and the second type as “general” influencers. Niche and general influencers’ followers respond differently to the endorsement of a randomly selected product and a well-matched product.

³Kuaishou is Tiktok’s biggest competitor in China. It is a Chinese video-sharing mobile app developed by Beijing Kuaishou Technology Co., Ltd, hosting over 200 million daily active users in 2019.

⁴Source: <https://www.shangyexinzhi.com/article/147644.html>. *Report of Kuaishou e-commerce marketing value (in Chinese)*.

⁵<https://medium.com/@talitarhein/niche-influencers-vs-general-influencers-7373cfa9f7c2>

⁶<https://izea.com/2020/01/09/lifestyle-influencers/>.

Our model makes the following predictions. First, when the product defective rate is sufficiently high on a platform, the platform will refrain from developing AI technology for matching influencers and marketers, even when the cost of AI development is negligible. Second, even if the product defective rate is not very high, a platform does not necessarily have the incentive to perfect its AI prediction accuracy. This happens when the proportion of loyal followers for influencers is not too large. On the one hand, a more accurate AI leads to a better match between the influencers and marketers and eventually higher sales. Thus, on-the-platform marketing generates more revenue for the influencers, which we refer to as the “sales effect.” On the other hand, influencers will lose more followers due to a better match if the products purchased by followers turn out to be defective, which we refer to as the “quality concern effect.” The trade-off between these two effects determines whether or not an influencer participates in on-the-platform marketing campaigns as the matching AI improves, which ultimately affects the platform’s revenue. Therefore, we could observe a non-linear relationship between the platform’s revenue and its AI accuracy. In particular, the proportional size of “loyal” followers moderates the trade-off between the above two effects, and thus serves as an important parameter a platform should consider when designing its AI strategy. We also investigate how a platform’s AI strategy will interact with its choice of service fee for influencer marketing. Interestingly, the platform would either improve or downgrade its AI prediction accuracy when they gain sufficiently high percentage of “loyal” followers, depending on whether the service fee is optimally set by the platform.

We also analyze two extensions to our baseline model to reflect more nuanced decisions in developing AI, including how the platform’s AI strategy will change if influencers need to incur a non-negligible opportunity cost of participating in on-the-platform marketing, and whether a platform should gather more information about product quality and communicate it to the influencers, given that product quality is a big concern.

Our study provides several managerial implications to social media platforms who aim to keep up with the industry trend of influencer marketing. First, although AI can facilitate matching influencers to marketers, developing such AI could backfire when the quality of products endorsed on the platform is low. Therefore, it is not always in a platform’s best interest to adopt this new AI technology. Second, given that quality concern is a source of negative impact from AI improvement, a quality assurance program could be necessary for a platform before it adopts and deploys such AI technology. Thus, regulating products on influencers’ recommendation channels is more than a legal issue and should be the policy consideration in regulating influencer marketing. Third, AI development could shift the

landscape of influencer marketing in that some types of influencers may opt in (or out of) on-the-platform marketing campaigns as AI improves. Finally, some platforms may be more willing to develop such AI because of their unique influencer-follower relationship, where influencers have a high percentage of loyal followers. This generates a testable hypothesis for the timing of AI adoption across different social media platforms. In an era of growing consumer privacy concerns, influencer marketing enables targeting without the need to collect private data. This targeting value of influencers makes “influencer placements especially valuable in cases where consumers or brands are sensitive to privacy violations (Campbell and Farrell, 2020, p. 473).” Our study is thus a timely discussion on how this new advertising model alters the shape of online activities.

We contribute to three strands of literature. First, our paper is related to the emerging research inspired by the growing interest in influencer marketing (Lou and Yuan, 2019; Pei and Mayzlin, 2019; Fainmesser and Galeotti, 2019; Berman and Zheng, 2020). Lou and Yuan (2019) proposes a conceptual model which investigates how characteristics of influencers will affect consumers’ trust in branded posts and brand awareness. Pei and Mayzlin (2019) show that in order to preserve the value of recommendations, marketers partially affiliate with influencers when the consumer’s prior belief on product fit is low, even if there is a possibility that influencers may post negative reviews on the product. Berman and Zheng (2020) study the impact of shallow influencers (who communicate to users whatever information the marketer provided) and prudent influencers (who test and try products of marketers before endorsement) on marketer profits, customer satisfaction, and influencer payoffs. None of the above papers, however, studies how the AI technology that reduces search frictions between marketers and influencers impacts the practice of influencer marketing.

The paper most closely related to ours is Fainmesser and Galeotti (2019), a theoretical study on how influencers trade-off endorsing paid products with creating organic content, which explains the phenomenon of the “rise of micro-influencers.” As a complement to our research, their paper also discusses the role of search technology, albeit the one matching followers to influencers, in shaping the competition among influencers and thus the welfare. On a social media platform with both influencers and users, an influencer can deliver content to users in two different pools, which is illustrated in Figure 1. The first one is called the “private user pool,” which consists of users who are followers of the influencer. For example, if a user subscribes to a YouTube channel, she is considered part of the “private user pool” of the YouTuber (influencer). In this pool, influencers “directly and frequently” interact with users because followers can consistently watch content generated by the same

influencer.⁷ The second one is called the “public user pool.” This pool includes other users, i.e., non-followers of the influencer, who access the influencer’s content via a search engine or algorithmic recommendations. Influencers only have limited interactions with users in this pool. In fact, delivering content to users in the “public user pool” is not a unique feature of influencer marketing, since this procedure is intrinsically similar to many other practices in digital marketing, such as Amazon’s product recommendation engine. In other words, Fainmesser and Galeotti (2019) focus on the search technology for influencers and users in the “public user pool.” However, engaging users in the “private user pool” is exclusive to the practice of influencer marketing, which is the focus of our paper.

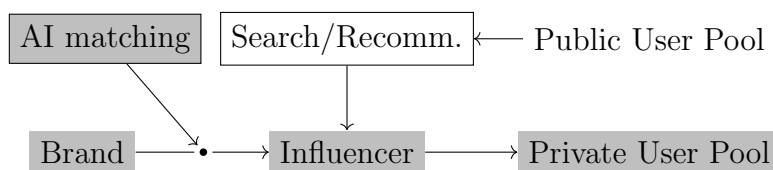


Figure 1: Influencer Marketing

Second, beyond our contribution to research in influencer marketing, we also contribute to a growing literature at the intersection of computer science, economics, and management science: search frictions in online markets. Search frictions in social media lead many platforms to develop curation algorithms to help users search for the most relevant content (in our case, influencers). Existing works have largely examined how to better design such algorithms matching users with influencers (Shardanand and Maes, 1995), or their effect on the content produced (see Latzer et al. (2016) for a survey). However, search frictions between brands and influencers are less studied, especially from a managerial perspective. Recent studies, mostly in computer science, begin to develop machine learning techniques for matching brands with social media influencers (Sweet et al., 2019). To the best of our knowledge, our paper is the first one that studies the economic incentives of designing such algorithms, and more importantly, its managerial implications.

Third, our paper also contributes to the research on the business implications of applying AI to marketing contexts. For example, Miklós-Thal and Tucker (2019) show that a more accurate demand prediction by AI may not always facilitate collusion between sellers and thus may decrease or increase consumer welfare. Lambrecht and Tucker (2019) empirically show that using an effectiveness-maximizing AI-algorithm in ad delivery may lead to gender-based discrimination. Luo et al. (2019) empirically document that despite the high performance of

⁷<https://www.campaignasia.com/article/move-over-kols-kocs-and-private-domain-traffic-are-hot-in-china/454639>.

AI-enabled chatbots, disclosing its machine identity reduces consumer purchases. Lee and Hosanagar (2019) claim that using collaborative filter-based AI recommenders may induce a decrease in market diversity. The above-mentioned papers show that the social and economic impact of AI improvement is not as straightforward as it seems. Our study contributes to this stream of literature by exploring the nuanced impact of AI improvement on firms’ profitability in the context of influencer marketing.

The rest of the paper is organized as follows. Section 2 describes the model. We provide our main analysis in Section 3. In Section 4, we make two extensions to our benchmark model to generate additional insights. Finally, we conclude in Section 5.

2 Model

2.1 The Influencers and Marketers

Consider a social media platform with mass one of influencers and sufficiently many marketers. An influencer’s payoff comes from two types of revenue-generating behavior: (1) on-the-platform marketing, and (2) off-the-platform business. In the following two subsections, we will discuss these two sources of revenue for influencers in detail.

On-the-platform marketing

The first revenue source of influencers is that they can earn a commission fee through paid endorsement on the platform, which we refer to as “on-the-platform marketing.” Influencers actively endorse a product/brand through live-streaming or well-prepared content such as short videos or articles. Within the influencer-generated content, there is usually a hyperlink which can direct followers to the purchase page of the endorsed product. As a result, a marketer can easily track the sales converted from an influencer’s engagement marketing campaign such as actively endorsing a product through live-streaming. Influencers are then compensated based on their sales performance. An influencer i will choose whether or not to actively endorse the product, depending on whether the potential of obtaining more profit outweighs the concern of losing followers, which we will discuss shortly. We denote this decision variable as $e_i \in \{0, 1\}$, where $e_i = 1$ if the influencer actively endorses a product, and otherwise $e_i = 0$.

Suppose that an influencer i has an initial follower base of size $z_i \in \mathbf{R}^+$. A percentage $\theta \in (0, 1)$ of her followers are “loyal” followers, while the remaining $1 - \theta$ are “non-loyal”

ones. “Loyal” followers would purchase any product endorsed by the influencer, regardless of whether they are interested in the product itself, simply to show their support. This reflects the distinct characteristic of influencer marketing mentioned in Section 1. “Non-loyal” followers, however, will buy the product if and only if they are interested in the product⁸. Without loss of generality, we assume each follower purchases at most one unit of the product. If the influencer i participates in on-the-platform marketing and endorses a product from marketer j , the expected number of her non-loyal followers who would buy the endorsed product is given by $z_i(1 - \theta)m_{i,j}$, where $m_{i,j} \in [0, 1]$ is the fraction of influencer i ’s non-loyal followers who are interested in the endorsed product from marketer j . In other words, $m_{i,j}$ is a *matching index* which captures the match between a marketer j and the followers of an influencer i . In the analysis that follows, we refer to the matching index $m_{i,j}$ as the matching between influencer i and marketer j . For simplicity and tractability, we assume that for each influencer i , there are two discrete levels of matching indices among all potential marketers, i.e., $m_{i,j} \in \{\overline{m}_i, \underline{m}_i\}$, where $\overline{m}_i > \underline{m}_i$. A marketer j is either “suitable” for influencer i ($m_{i,j} = \overline{m}_i$) if a *high* percentage of influencer i ’s non-loyal followers⁹ are interested in the product from marketer j , or “unsuitable” for her ($m_{i,j} = \underline{m}_i$) if a *low* percentage of her non-loyal followers are interested in the product from marketer j . In practice, for example, influencers on a major social media platform in China are matched to marketers based on overlapping metrics between the characteristics of influencers’ followers and the user portrait specified by the marketers.¹⁰

We assume that the quality of advertised products on the platform, q , can be either L (low) or H (high), with probability ω and $1 - \omega$, respectively. $\omega \in [0, 1]$ indicates the defective rate among the products on the platform. The influencers have a rational belief about the product quality distribution on the platform, but cannot observe the quality of a specific product¹¹. A user will stop following the influencer if and only if she is non-loyal and has purchased a low-quality product via the influencer’s recommendation channel. If the influencer does not participate in on-the-platform marketing, no followers will stop following the influencer. Let $\hat{z}_i$ denote influencer i ’s follower base after on-the-platform marketing.

⁸For simplicity, we assume that a follower would purchase the product as long as s/he is interested. This assumption is innocuous to our qualitative results.

⁹When defining the matching index, we do not consider the interest of loyal followers because they will purchase any product endorsed by the influencer regardless of their own preferences for the product itself. Thus their purchase decisions are independent of the improvement of brand-influencer matching.

¹⁰This is confirmed by our in-depth interview with one of the managers at this company who is in charge of its on-the-platform influencer marketing campaigns.

¹¹Even if the influencer receives and tests the sample provided by the brand before endorsing its product, she can never be sure or fully control whether the products delivered to her followers will be of the same quality as the sample. Source: <https://www.yicai.com/news/100685877.html>.

Depending on the influencer's participation decision, $\hat{z}_i$ may be different from z_i .

Influencer i receives a commission from the marketer if she participates in on-the-platform marketing. The commission is proportional to the sales (S_i) that she induces, which equals αS_i , where $0 \leq \alpha \leq 1$. In our model, we assume that all products have the same price and normalize it to 1 for simplicity. Since the platform observes and regulates on-the-platform marketing campaigns, influencers who participate in on-the-platform marketing must pay a proportion of $0 \leq \delta \leq 1$ from her commission as service fee to the platform.

In sum, the sales that an influencer i converts from on-the-marketing campaigns, S_i , is determined by the influencer's participation decision (e_i) and the matching index between the influencer and the marketer ($m_{i,j}$). The number of followers she retains after on-the-platform marketing, $\hat{z}_i$, is additionally determined by the realized quality of the product (q). Their expressions are specified in the following table (Table 1), given that the matching index is m .

Table 1: Induced sales S_i and number of followers after on-the-platform marketing $\hat{z}_i$

Influencer i 's decision (e_i)	Product quality (q)	Low (L)	High (H)
Participate ($e_i = 1$)		$S_i(e_i = 1; m) = z_i\theta + z_i m(1 - \theta),$ $\hat{z}_i(e_i = 1, L; m) = z_i - z_i m(1 - \theta)$	$S_i(e_i = 1; m) = z_i\theta + z_i m(1 - \theta),$ $\hat{z}_i(e_i = 1, H; m) = z_i$
Not participate ($e_i = 0$)		$S_i(e_i = 0; m) = 0,$ $\hat{z}_i(e_i = 0, L; m) = z_i$	$S_i(e_i = 0; m) = 0,$ $\hat{z}_i(e_i = 0, H; m) = z_i$

In reality, some influencers tend to gain their following in a specific niche by creating content relevant to that niche, whereas others have a diverse follower base.¹² We model this phenomenon by assuming that there are two types of influencers $i \in \{N, G\}$: (1) “niche” influencers (N) who create content focused on a cluster of subjects that usually attract like-minded followers with particular interests, and (2) “general” influencers (G) who are versatile and do not focus on one topic. General influencers are sometimes also referred to as “lifestyle influencers” who usually have a diverse follower base. To capture the difference between the two types of influencers, we assume that $\overline{m}_N > \overline{m}_G$ and $\frac{1}{2}(\overline{m}_N + \underline{m}_N) < \frac{1}{2}(\overline{m}_G + \underline{m}_G)$. That is, niche influencers have proportionally more followers who are interested in a suitable product than general influencers; on the other hand, niche influencers have proportionally fewer

¹²<https://medium.com/@talitarhein/niche-influencers-vs-general-influencers-7373cfa9f7c2>.

followers who will be interested in a randomly picked¹³ product than general influencers, since the followers of niche influencers have more concentrated characteristics. Note that general influencers can be either micro or big. In other words, z_G can be either larger or smaller than z_N .¹⁴ Let ρ_N and ρ_G denote the number of niche and general influencers, respectively, in the market. Let $\eta \equiv \frac{\rho_N z_N}{\rho_G z_G}$, which indicates the ratio between the total number of users who follow niche influencers and general influencers. In other words, niche (general) influencers dominate the market if η is larger (smaller) than 1.

Therefore, an influencer i obtains a net revenue of π_i^{on} from on-the-platform marketing campaigns, given by

$$\pi_i^{on}(e_i; m) = (1 - \delta)\alpha S_i(e_i; m), \quad (1)$$

where $S_i(e_i; m)$ is the number of purchases induced by influencer i 's recommendation channel.

Off-the-platform business

Besides on-the-platform marketing, influencers also pursue other opportunities to generate income. Some have collaborated with brands to “co-produce new product lines or brands,”¹⁵ while others “converted their influence into supporting their own entrepreneurial ventures (Avery and Israeli, 2020).” Their followers contribute through Patreon or subscribe to services and products offered by influencers’ own business ventures. While the revenue from this type of business is closely linked to an influencer’s follower size, the platform cannot charge service fees because it is not involved in the business. Hence we refer to it as the “off-the-platform” business.

To highlight the trade-off between the two types of revenue-generating behavior, we intentionally assume that there are two periods in the game: off-the-platform business happens in both periods, while on-the-platform marketing campaign occurs only in the first period. The revenue from Period 2 represents all the future value from off-the-platform businesses of an influencer.

We assume that influencers earn a revenue that is proportional to the number of their followers from off-the-platform businesses in both periods. Specifically, an influencer obtains

¹³We assume the probability that a “suitable” marketer is picked for an influencer is the same as the probability that an “unsuitable” one is picked (and thus is $\frac{1}{2}$), if the choice is totally at random.

¹⁴For ease of communication, we assume the same type of influencers have the same number of followers. Relaxing this assumption will not alter any of our qualitative results.

¹⁵For example, Gucci took advantage of the famous YouTube vlogger Jeffree Star’s love for Gucci and ended up designing a custom “Pink Luggage” collection exclusively for him. Millions of his followers read about this gesture and earned Gucci not just media mentions but also sales worth \$1.6 million.

a revenue of $\pi_i^{off}(e_i, q; m)$ from off-the-platform business, given by

$$\pi_i^{off}(e_i, q; m) = \beta z_i + \beta \hat{z}_i(e_i, q; m), \quad (2)$$

where $\beta > 0$ is the average revenue per user (ARPU) in each period.

The Decision of the Influencers

Since an influencer can earn revenue from both on-the-platform marketing and off-the-platform businesses, we can write influencer i 's total revenue function as $\pi_i = \pi_i^{on} + \pi_i^{off}$, where π_i^{on} is the payoff from on-the-platform marketing, and π_i^{off} is the compensation from off-the-platform businesses, defined before. We assume that at the time when an influencer is making her participation decision, she does not have perfect knowledge about whether the product is "suitable" for her or not. Moreover, quality (q) is not observable to influencers. Thus, an influencer i 's expected revenue function is given by

$$\mathbf{E} \pi_i(e_i) = \mathbf{E}_{q,m}[\pi_i^{on}(e_i; m) + \pi_i^{off}(e_i, q; m)].$$

In particular, if the influencer chooses to participate in on-the-platform marketing campaigns, her expected payoff is

$$\mathbf{E} \pi_i(e_i = 1) = \underbrace{(1 - \delta)\alpha z_i(\theta + (1 - \theta) \mathbf{E}(m))}_{\text{payoff from on-the-platform marketing}} + \underbrace{\beta z_i + \beta(z_i - \omega z_i(1 - \theta) \mathbf{E}(m))}_{\text{payoff from off-the-platform businesses}}; \quad (3)$$

if she chooses not to participate, her expected payoff is

$$\mathbf{E} \pi_i(e_i = 0) = 2\beta z_i. \quad (4)$$

As a tie-breaking rule, we assume that an influencer who is indifferent between endorsing a product and not endorsing will choose to endorse it. Therefore, influencer i participates in on-the-platform marketing iff $\mathbf{E} \pi_i(e_i = 1) \geq \mathbf{E} \pi_i(e_i = 0)$.

2.2 The Platform

For on-the-platform marketing, the platform has the option of offering an AI-based service that increases the probability that marketers are matched with "suitable" influencers. Such a service is viable because the platform not only has access to influencers' metadata but

also the historical sales data generated by their marketing campaigns, and thus can build an AI/machine learning model to help match marketers with influencers. Mathematically, we model that the platform's AI-based service can match an influencer i with a suitable marketer ($m = \overline{m}_i$) with probability $\frac{1}{2} + \kappa$, and with an unsuitable marketer ($m = \underline{m}_i$) with probability $\frac{1}{2} - \kappa$, where $\kappa \in [0, \frac{1}{2}]$. Here, the parameter κ measures how accurate the AI matching algorithm is: when $\kappa = 0$, the AI randomly matches an influencer with a suitable or unsuitable marketer with equal probability, which is equivalent to the case of no AI; when $\kappa = \frac{1}{2}$, the AI can match an influencer with a suitable marketer with probability 1.

The platform charges a percentage service fee for providing the marketplace to influencers and marketers for such on-the-platform marketing campaigns, which is considered to be the major source of the platform's revenue.¹⁶ To highlight the misalignment between the platform and the influencers in their technology preferences, which we will discuss soon, we assume that the cost of developing AI technology is negligible.¹⁷ The platform's expected revenue function, Π , given the influencers' participation decisions $e = (e_G, e_N)$, is characterized by

$$\Pi(e) = \mathbf{E}_m \sum_i \delta \alpha \rho_i S_i(e_i; m). \quad (5)$$

The platform chooses the level of AI precision κ to maximize its expected revenue¹⁸.

2.3 Timing and Equilibrium Concept

The timing of the game is characterized as follows:

- The platform first chooses the level of AI prediction accuracy $\kappa \in [0, \frac{1}{2}]$ and broadcasts it to the public.
- In Period 1,
 1. each influencer is matched with a marketer and chooses whether to participate in on-the-platform marketing (e_i);
 2. the platform and each influencer share the revenues generated by on-the-platform marketing campaigns;

¹⁶In the benchmark analysis, we assume that this is a fixed percentage fee out of the commission a marketer pays to an influencer, and it is independent of the development of AI. In Section 3.1, we relax this assumption and allow the platform to adjust δ as it chooses whether or not to develop AI.

¹⁷A non-negligible cost of developing AI will be more supportive for our main argument in Section 3.

¹⁸As a tie-breaking rule, we assume that the platform will choose the minimum level of prediction accuracy κ that yields the maximum profit. For example, if the platform is indifferent between developing and not developing AI, it will choose not to develop AI, i.e. $\kappa = 0$.

3. meanwhile, influencers obtain revenue from their off-the-platform businesses.
- In Period 2,
 1. followers who have purchased products through the influencers' recommendation channel discover the true quality of an endorsed product, and non-loyal followers stop following an influencer if they purchased a low-quality product;
 2. influencers again obtain revenue from their off-the-platform businesses.

Our solution concept is subgame perfect equilibrium. We start with studying the benchmark case where the percentage commission fee (δ) is fixed and there is no information asymmetry between the platform and the influencers. That is, the platform and the influencers share a common belief about product quality. We then analyze the case where the platform can simultaneously adjust the commission fee as it develops AI technology in Section 3.1. Finally, we examine the case where the platform has more information about product quality than the influencers in Section 4.2. In the next section, we provide the solution and analysis of the benchmark model. Table 2 summarizes the key notations in our model.

Table 2: Key Notations in the Model

Notation	Range	Definition
e_i	$\in \{0, 1\}$	influencer i 's endorsement decision
z_i	$\in \mathbf{R}^+$	size of influencer i 's follower base
θ	$\in (0, 1)$	percentage of loyal followers
$m_{i,j}$	$\in \{m_i, \overline{m}_i\}$	matching index between influencer i and marketer j
$\underline{m}_i$	$\in [0, \overline{m}_i]$	matching index when a marketer is <i>not</i> suitable for influencer i
$\overline{m}_i$	$\in [m_i, 1]$	matching index when a marketer is suitable for influencer i
ω	$\in (0, 1]$	product defective rate
α	$\in (0, 1]$	influencers' percentage commission fee
δ	$\in (0, 1]$	percentage service fee charged by the platform
η	$\in \mathbf{R}^+$	ratio between the total number of niche and general followers
β	$\in [0, 1]$	influencers' revenue per user from off-the-platform businesses
κ	$\in [0, \frac{1}{2}]$	prediction accuracy of AI

3 Analysis

Given the choice of the platform, we first investigate how influencers makes their participation decision.

Lemma 1. *For a given level of AI technology κ , influencer i participates in on-the-platform marketing, i.e., $e_i^* = 1$, if and only if $\omega \leq \bar{\omega}_i(\kappa)$ where*

$$\bar{\omega}_i(\kappa) \equiv \frac{(1-\delta)\alpha\theta}{\beta(1-\theta)(\bar{m}_i(\frac{1}{2}+\kappa) + \underline{m}_i(\frac{1}{2}-\kappa))} + \frac{(1-\delta)\alpha}{\beta}. \quad (6)$$

Lemma 1 indicates that influencers participate in on-the-platform marketing only when they believe that the defective rate of endorsed products on the platform is lower than a threshold value. Otherwise, they would be better off not participating because the future risk of losing followers outweighs the current revenue gain from on-the-platform marketing campaigns.

Note that, if $(1-\delta)\alpha \geq \beta$, the incentives of the platform and the influencers are always aligned regardless of other parameters including the chosen level of AI prediction accuracy. Hence in the analysis that follows, we focus on the non-trivial cases by making the following assumption:

$$(1-\delta)\alpha < \beta. \quad (7)$$

With some algebra, it is not hard to see that $\bar{\omega}_i(\kappa)$ is decreasing in κ . That is, if there are enough loyal followers, influencers' quality concern increases with the accuracy of the matching technology. In other words, influencers are more conservative about quality as AI accuracy improves. Furthermore, we can see that in this case, there exists a $\hat{\kappa} \equiv \frac{(\bar{m}_G + m_G) - (\bar{m}_N + m_N)}{2[(\bar{m}_N - \underline{m}_N) - (\bar{m}_G - m_G)]}$ such that $\bar{\omega}_G(\kappa) < \bar{\omega}_N(\kappa)$ when $\kappa < \hat{\kappa}$ and $\bar{\omega}_G(\kappa) > \bar{\omega}_N(\kappa)$ when $\kappa > \hat{\kappa}$, which is depicted in Figure 2. That is, general influencers are more (less) conservative about the quality of products than niche influencers when the technology is inaccurate (accurate).

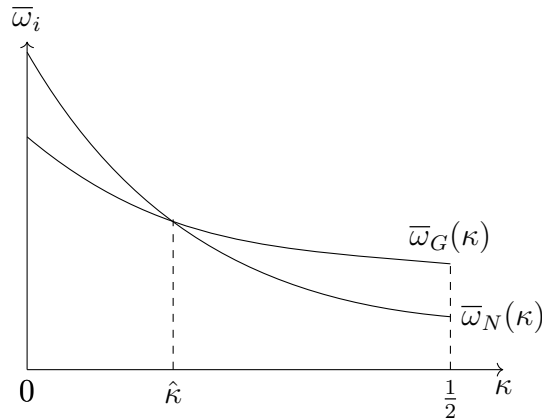


Figure 2: Shape of $\bar{\omega}_i(\kappa)$

For any defective rate ω , we can derive the choice of each influencer based on Lemma

1. Then we calculate the platform's profit based on equation (5). Taking into account the decisions of influencers, the platform chooses the optimal level of AI technology to maximize its revenue characterized by equation (5). The following proposition describes the platform's equilibrium technology choice (κ^*), which is also illustrated by Figure 3.

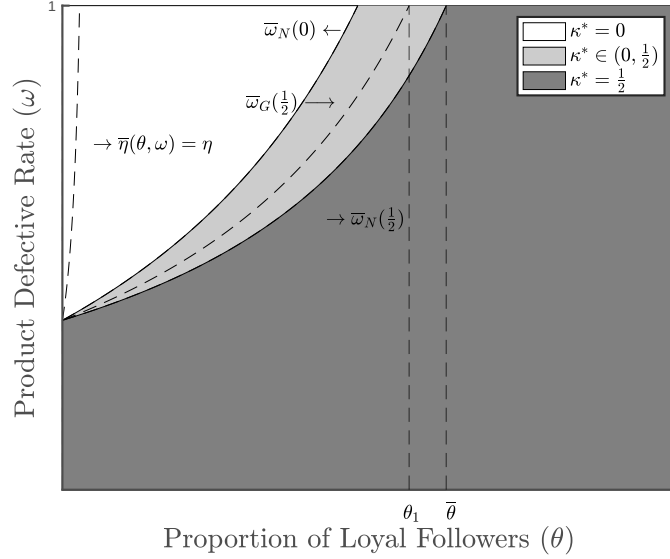


Figure 3: Optimal Choice of AI (κ^*)

Note: $\delta = 0.3, \alpha = 0.5, \beta = 1, \bar{m}_N - \underline{m}_N = 0.8, \bar{m}_G - \underline{m}_G = 0.2, \eta = 5$
 $^*\eta(\cdot)$ is an increasing function of ω .

Proposition 1. (Optimal Choice of AI) *In equilibrium, the platform chooses*
(i) not to develop AI if the defective rate is sufficiently high. In other words,

$$\omega > \bar{\omega}_N(\kappa = 0) \implies \kappa^* = 0;$$

(ii) to develop a “less-than-perfect” AI if influencers do not have a large proportion of loyal followers and the defective rate is moderately high. In other words,

$$\theta < \bar{\theta} \text{ and } \begin{cases} \bar{\omega}_G(\kappa = \frac{1}{2}) < \omega < \bar{\omega}_N(\kappa = 0) & \text{if } \theta < \theta_1 \\ \bar{\omega}_N(\kappa = \frac{1}{2}) < \omega \leq \bar{\omega}_G(\kappa = \frac{1}{2}), \eta > \bar{\eta}(\theta, \omega) & \text{if } \theta_1 \leq \theta < \bar{\theta} \end{cases} \implies \kappa^* \in (0, \frac{1}{2});$$

(iii) to develop a “perfect” AI (i.e. $\kappa^ = \frac{1}{2}$) if otherwise.*

The expressions for θ_1 , $\bar{\theta}$, and $\bar{\eta}(\theta, \omega)$ are provided in the Appendix on page A5. Proposition 1 and Figure 3 summarize the platform's optimal choice of AI technology for matching

marketers with influencers under different circumstances. Interestingly, the platform may refrain from developing AI technology even if the cost of developing AI is negligible. Specifically, a platform is less likely to develop AI when ω is sufficiently high, that is, when influencers hold a pessimistic belief about the quality of the endorsed products. To provide some intuition, we can decompose the total impact of improving AI into the following two effects:

- *Sales effect.* Sales induced by on-the-platform marketing campaigns is higher when marketers (products) are better matched with influencers.
- *Quality concern effect.* For a given product defective rate, influencers are expected to lose more non-loyal followers when they are better matched with marketers (products).

The sales effect is positive to both the influencers' and the platform's revenue as their revenue is proportional to sales. However, the quality concern effect is negative to the influencers' revenue as a better AI reduces their expected future revenue which is proportional to their follower size, from both on-the-platform and off-the-platform businesses. On the other hand, the quality concern effect does not apply to the platform because its revenue depends only on the sales from on-the-platform marketing.¹⁹ When the defective rate of products is sufficiently high, this asymmetry leads to severe misalignment between the incentives of platform and that of influencers when it comes to AI improvement. In those cases, the platform has to give up completely on the AI development ($\kappa^* = 0$) just to induce more influencers to participate in the marketing campaigns. This result is consistent with the anecdotal evidence that Kuaishou, one of China's largest influencer-marketing platforms, announced a quality insurance program before it launched its AI system for matching marketers and influencers²⁰.

Proposition 1 also suggests that, even when the platform chooses to develop AI, it does not always have an incentive to perfect its matching algorithm. In particular, if both the product defective rate (ω) and the proportion of influencers' loyal followers (θ) are not too high, the platform may purposefully cap its AI prediction accuracy below the maximum level. When θ is sufficiently large ($\theta > \bar{\theta}$), that is, when influencers have a lot of loyal followers, the platform's optimal technology choice is independent of product quality: a better AI always benefits the platform. This is because loyal followers never leave the influencers regardless of the realized product quality, which alleviates the quality concern effect on influencers.

¹⁹Note that the platform may also earn revenue directly from its users (e.g., through display ads), but our results remain unchanged unless users who stop following an influencer leave the platform. In that case, the technology preference misalignment between the platform and influencers will be alleviated.

²⁰<https://tech.huanqiu.com/article/9CaKrnK1fPV>.

Thus, sales effect is more likely to dominate in this case. In the early stage of a platform, most influencers tend to have very limited, if any, loyal followers. As the platform starts to gain popularity among users, the percentage of influencers' loyal follower base also tends to grow. Once again, anecdotal evidence seems to corroborate our hypothesis that a platform has the incentive to improve its AI only when its influencers have accumulated a significant proportion of loyal followers.

Our results so far are intuitive: as long as influencers do not have too many loyal followers, the platform's optimal technology choice is monotonically related to the product defective rate. Specifically, it is possible that the platform chooses to develop AI but intentionally caps its AI's prediction accuracy, for any given $\theta < \bar{\theta}$. This is because so far we have assumed away the opportunity cost of on-the-platform marketing. In reality, influencers will incur some opportunity cost when they devote their time to participating in on-the-platform marketing campaigns because it prevents them from engaging in other profit-generating businesses. In Section 4.1, we will relax this assumption and explore the case where this opportunity cost is not negligible. We will show in that case, platform may never want to develop a less-than-perfect AI even when the influencers have very few loyal followers.

Given the optimal technology choice characterized by Proposition 1, we then examine the equilibrium decision of different types of influencers in the market and summarize the results in Corollary 1.

Corollary 1. (Who Participates in on-the-Platform Marketing) *If the niche influencers participate in on-the-platform marketing in equilibrium, then the general influencers must also participate (i.e. $e_N^* = 1 \implies e_G^* = 1$), unless one of the following two conditions are satisfied:*

- (1) $\omega \geq \bar{\omega}_G(\kappa = 0)$ or
- (2.1) $\bar{\omega}_G(\kappa = \hat{\kappa}) < \omega < \bar{\omega}_G(\kappa = 0)$ and (2.2) $\eta > \bar{\eta}_1(\theta, \omega)$, where $\bar{\eta}_1(\theta, \omega) > 1$.

The expression for $\bar{\eta}_1(\theta, \omega)$ is provided in Appendix on page A6. As shown in Figure 2, the defective rate threshold ($\bar{\omega}_i$) decreases as the matching technology improves, i.e., $\frac{\partial \bar{\omega}_i}{\partial \kappa} < 0, i = G, N$. Furthermore, we see that the general influencers have a flatter threshold function than niche ones ($|\frac{\partial \bar{\omega}_G}{\partial \kappa}| < |\frac{\partial \bar{\omega}_N}{\partial \kappa}|$). That is, general influencers are less affected by the improvement of AI accuracy than niche ones. If condition (1) in Corollary 1 holds, then general influencers would not participate in on-the-platform marketing regardless of the AI prediction accuracy. Niche influencers participate whereas general ones do not only if the former's decisions are less sensitive to product quality than the latter's decisions ($\bar{\omega}_N > \bar{\omega}_G$),

which occurs when the equilibrium AI prediction accuracy is low enough ($\kappa^* < \hat{\kappa}$). This happens only when the defective rate is higher than $\bar{\omega}_G(\kappa = \hat{\kappa})$ (condition (2.1) in Corollary 1). Note that under condition (2.1), there still remain two possible scenarios: both types of influencers participate or only niche influencers participate. Which scenario appears in equilibrium is subject to the platform’s choice. The platform earns more revenue from the latter scenario only when niche influencers dominate the market in terms of their total number of followers, which is condition (2.2) in Corollary 1.

Results in Corollary 1 could serve as a testable hypothesis generated by our theoretical analysis. To provide some *prima facie* evidence, we have collected data from a major Chinese social media platform that conducts influencer marketing campaigns. The dataset includes the metadata of Top 1000 influencers on the platform, among which we focus on 1) the number of followers of each influencer and 2) information that indicates whether or not they have participated in on-the-platform marketing campaigns (see Appendix B on page A15 for details). To identify the type (i.e., general/niche) of each influencer, we employed several independent coders to rate the influencer on a scale of 1 to 5 based on the video content generated by the influencer. We define an influencer as “niche” if the average rating is less than or equal to 3, and otherwise as “general.”²¹

We assume that the product defective rate is not so high that it becomes impossible for general influencers to participate in on-the-platform marketing under any AI environment, which precludes condition (1) in Corollary 1. Our empirical data shows that the ratio between the total number of users who follow niche influencers and general influencers $\eta = 0.42 < 1$, which violates condition (2.2) in Corollary 1. Therefore, Corollary 1 predicts that general influencers are more likely to participate in on-the-platform marketing than niche ones. Our empirical results confirm this prediction. We refer readers who are interested in the detailed empirical analysis to Appendix B on Page A15.

3.1 Endogenous Commission Rate

So far, we have assumed that the revenue generated from on-the-platform marketing is split between an influencer and the platform based on an exogenous percentage service fee δ . In reality, this is generally true in the short term. In the long run, however, the platform can make a decision on the percentage fee it wants to charge from the campaign revenue. In other words, $\delta \in [0, 1]$ can be endogenously set by the platform. We discuss such possibility in this

²¹Our results are robust to an alternative classification scheme which classifies influencers with a rating of “3” as “general” ones.

subsection and see how it would affect the platform's choice of AI matching technology.

If influencers do participate in on-the-platform marketing, the platform always prefer a higher percentage service fee (δ). On the other hand, a too large δ will not provide enough incentive for influencers to participate. Therefore, the platform needs to optimally adjust its service fee to balance its revenue share against influencers' participation constraints. Note that in previous sections, we assumed $(1 - \delta)\alpha < \beta$ which precludes trivial cases. Now that δ is endogenous, we modify this assumption by assuming it holds even when the platform sets a zero service fee, i.e., $\delta = 0$. That is, we assume:

$$\alpha < \beta. \quad (8)$$

Lemma 2. *For any given κ , the optimal percentage service fee δ^* is either δ_G or δ_N , whichever gives the higher profit to the platform, where*

$$\delta_i = \max \left\{ 0, 1 - \frac{\beta\omega(1 - \theta) (\overline{m}_i (\frac{1}{2} + \kappa) + \underline{m}_i (\frac{1}{2} - \kappa))}{\alpha (\theta + (1 - \theta) (\overline{m}_i (\frac{1}{2} + \kappa) + \underline{m}_i (\frac{1}{2} - \kappa)))} \right\} \quad (9)$$

for $i = G, N$.

Note that $\delta^* = 0$ indicates the case where even if the platform does not charge any service fee, the influencers will not participate in on-the-platform marketing. This can happen if the expected quality of the products is sufficiently low (ω sufficiently large) and the influencers do not have enough "loyal" followers (θ is sufficiently small). Other than that, the platform would always set the service fee at a level where at least one type of influencers is indifferent between whether or not to participate in on-the-platform marketing. However, the platform still needs to choose between setting a lower fee such that all influencers participate and setting a higher fee but only some of the influencers do so. In fact, the platform's decision on the service fee also interacts with its decision on the prediction accuracy of the matching technology. The following proposition summarizes the optimal technology choice of the platform.

Proposition 2. (Optimal Choice of AI with Endogenous Service Fee) *If the platform can simultaneously choose its AI prediction accuracy and percentage service fee,*

1. *there still exists a region of (ω, θ) where the optimal technology is less-than-perfect ($\kappa^* \in (0, \frac{1}{2})$), and*

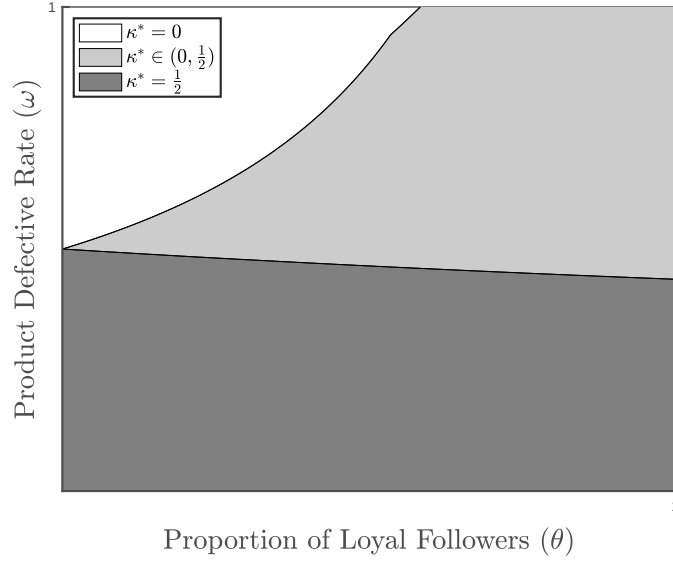


Figure 4: Optimal Choice of AI with Endogenous Service Fee (κ^*)

Note: $\alpha = 0.5, \beta = 1, \overline{m}_N - \underline{m}_N = 0.8, \overline{m}_G - \underline{m}_G = 0.2, \eta = 5$

2. *unlike in the benchmark model, the optimal technology can be less-than-perfect even when θ is sufficiently large ($\theta > \bar{\theta}$ in Proposition 1).*

Proposition 2 first suggests that our findings about the optimal technology choice remain relatively stable if the service fee is endogenously chosen by the platform: the optimal AI technology that matches influencers and advertisers may not be perfectly accurate. Furthermore, comparing Propositions 2 and 1 reveals that if, the platform can endogenously set its service fee, the optimal AI can still be less-than-perfect even when influencers have a large proportion of loyal followers. This would never occur if the service fee was exogenously given, where influencers will prefer to participate in on-the-platform marketing regardless of the AI prediction accuracy as long as they have sufficiently many loyal followers. As a result, when the service fee is exogenously given, the platform always prefers a perfect AI that generates more revenue when θ is sufficiently large ($\theta > \bar{\theta}$ in Proposition 1). On the other hand, if the service fee is also a decision variable of the platform, it can extract more surplus from influencers by increasing the service fee as influencers accumulate more loyal followers (i.e. larger θ). Meanwhile, however, higher service fees may discourage some influencers from participating in the campaign. To maintain a certain level of participation, the platform has to lower its AI prediction accuracy as a compromise for higher revenue.

4 Extensions

4.1 Opportunity Cost of on-the-platform Marketing

In the benchmark model, we have abstracted away from any opportunity cost associated with on-the-platform marketing campaigns. In reality, influencers may need to devote a significant amount of time into running an on-the-platform marketing campaign, especially when they proactively endorse products via creating short videos or live-streaming. For example, a famous Chinese influencer Li Jiaqi, also known as “the No.1 seller of lipstick” with an annual income of more than \$1.4 million (CNY 10 million), is said to have spent on average 9 hours a day²² in preparing and managing on-the-platform live-stream marketing campaigns. In other words, influencers have to incur an opportunity cost from participating in on-the-platform marketing because it may prevent them from engaging in and thus obtaining revenue from other off-the-platform businesses. In this section, we explore how the existence of such opportunity costs will affect the platform’s technology choice.

Specifically, we assume that the influencers earn less revenue from off-the-platform businesses in Period 1 if they participate in on-the-platform marketing in the same period than if they do not. In other words, conditional on her choice of whether or not to participate in on-the-platform marketing ($e_i = 1$ or 0), an influencer obtains a revenue of $\pi_i^{on}(e_i, q; m)$ (given by (1)) from on-the-platform marketing and a revenue of $\pi_i^{off}(e_i, q; m)$ given by

$$\pi_i^{off}(e_i, q; m) = \underbrace{(\beta - \tilde{\beta}e_i)z_i}_{\text{payoff in Period 1}} + \underbrace{\beta\hat{z}_i(e_i, q; m)}_{\text{payoff in Period 2}}, \quad (10)$$

from off-the-platform businesses. In equation (10), $\tilde{\beta} > 0$ indicates the opportunity cost of on-the-platform marketing in Period 1, q the product quality, and m the matching index introduced on page 7. In fact, Equation (10) is a generalized form of Equation (2).

Proposition 3. (Non-zero Opportunity Cost) *When the opportunity cost of on-the-platform marketing is sufficiently large for influencers (i.e. $\tilde{\beta} > \bar{\beta}^{EXO}$ in the exogenous service fee case or $\tilde{\beta} > \bar{\beta}^{END}$ in the endogenous service fee case) or when influencers have sufficiently small proportion of loyal followers (i.e. $\theta \leq \underline{\theta}^{EXO}$ in the exogenous service fee case or $\theta \leq \underline{\theta}^{END}$ in the endogenous service fee case), the optimal technology choice for the platform is limited to only two options: either a perfect AI ($\kappa^* = \frac{1}{2}$) when the expected product quality is high or otherwise no AI ($\kappa^* = 0$). Otherwise, our results remain qualitatively the*

²²<https://new.qq.com/omn/20191114/20191114A05LDW00.html>.

same as in Propositions 1 and 2.

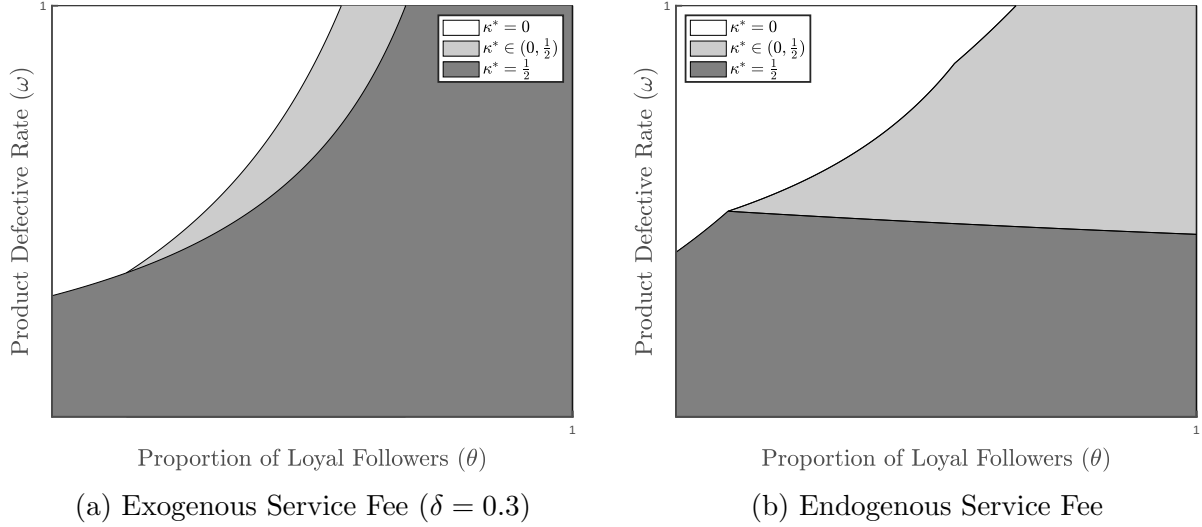


Figure 5: Optimal Choice of AI

Note: $\alpha = 0.5, \beta = 1, \tilde{\beta} = 0.05, \overline{m}_N - \underline{m}_N = 0.8, \overline{m}_G - \underline{m}_G = 0.2, \eta = 5$

As a comparison to Figures 3 and 4 in our benchmark model, Figure 5 illustrates the platform's optimal technology choice when influencers need to incur some opportunity cost from on-the-platform marketing campaigns ($\tilde{\beta} > 0$) but the cost is not too large (i.e. $\tilde{\beta} \leq \bar{\beta}^{EXO}$ (or $\bar{\beta}^{END}$)). First, when $\theta > \underline{\theta}^{EXO}$ (or $\underline{\theta}^{END}$), the pattern is similar to that in Figures 3 and 4. Interestingly, when influencers have sufficiently small fraction of loyal followers (i.e. $\theta \leq \underline{\theta}^{EXO}$ (or $\underline{\theta}^{END}$)), the platform prefers to match influencers to marketers as accurately as possible ($\kappa^* = \frac{1}{2}$) if it decides to develop AI at all. This is because when influencers have too few loyal followers, the opportunity cost of endorsing an ill-matched product is prohibitively high. In that case, when endorsing a product, the influencer could generate sales only from her followers who find the product matching their interests (as opposed to blind support). Hence influencers are more concerned about poor sales from on-the-platform marketing campaigns if the product is an ill match, as opposed to losing followers from endorsing the product if it turns out to be low-quality. In other words, sales effect dominates the quality concern effect here. As a result, the platform is more in favor of providing a better AI for matching influencers with marketers.

4.2 Product Quality (q) Is Private Information of the Platform

In the benchmark model, we assumed that the platform and all influencers share a common belief about the distribution of product quality, i.e., a product is low-quality with probability ω , but none of them can observe the true quality of product advertised on the platform. Because of their uncertainty about product quality, the platform has to trade-off a better AI technology with incentivizing the influencers to participate in on-the-platform marketing campaigns especially when low-quality products are prevailing. Given this undesirable situation, a natural question arises: if a platform has the resources to garble more information (e.g. data on past sales and refund rate) about a brand's product quality, will the platform communicate it to the influencers and if so, how? In this section, we explore this question by explicitly modeling the communication of the product quality between the platform and influencers as a Bayesian persuasion problem (Kamenica and Gentzkow, 2011).

Specifically, we assume that once a brand reaches out for collaboration with influencer i (either through platform's AI matching algorithm if available or by itself if not), the platform can acquire information from the brand and observe its true product quality. Upon observing the true product quality $q \in \{H, L\}$, the platform matches the brand with influencer i and sends a private signal $\hat{q} \in \{H, L\}$ to her that partially reveals q . In particular, the platform designs an information environment where the influencer i will receive a signal of "high-quality" ($\hat{q} = H$) with probability r_i if the product is indeed high-quality, and with probability s_i if the product is low-quality. Otherwise, the influencer receives a signal of "low-quality" ($\hat{q} = L$). The formal communication rule is given below:

$$\begin{aligned}\mathbf{P}_i(\hat{q} = H|q = H) &= r_i, \\ \mathbf{P}_i(\hat{q} = L|q = H) &= 1 - r_i, \\ \mathbf{P}_i(\hat{q} = H|q = L) &= s_i, \\ \mathbf{P}_i(\hat{q} = L|q = L) &= 1 - s_i,\end{aligned}$$

where $0 \leq r_i \leq 1, 0 \leq s_i \leq 1$. When the influencer i observes the signal $\hat{q}$, she makes the participation decision based on her updated belief about the product quality. By Bayes' Rule, the posterior probability of the product being low-quality conditional on receiving a

signal $\hat{q}$ is given by:

$$\begin{aligned}\mathbf{P}_i(q = L|\hat{q} = H) &= \frac{s_i\omega}{r_i(1-\omega) + s_i\omega}, \\ \mathbf{P}_i(q = L|\hat{q} = L) &= \frac{(1-s_i)\omega}{(1-r_i)(1-\omega) + (1-s_i)\omega}.\end{aligned}$$

By Lemma 1, we know that the influencer will participate in the marketing campaign ($e_i = 1$) upon receiving $\hat{q} = H$ if $\mathbf{P}_i(q = L|\hat{q} = H) \leq \bar{\omega}_i$ and upon receiving $\hat{q} = L$ if $\mathbf{P}_i(q = L|\hat{q} = L) \leq \bar{\omega}_i$. With some simple algebra, we can prove that the influencer cannot rationally participate after receiving both H and L signals. Also note that if we re-label the signal of high-quality to L and the signal of low-quality to H , the results are symmetric. Thus, without loss of generality, we assume that the influencer i will participate only upon receiving a message indicating high-quality $\hat{q} = H$.

Lemma 3. (Optimal Information Disclosure) *For any given technology $\kappa \in [0, \frac{1}{2}]$,*

(i) when the product defective rate is low ($\omega \leq \bar{\omega}_i$), the platform does not reveal product quality to influencer i .

(ii) when the product defective rate is high ($\omega > \bar{\omega}_i$), the platform truthfully reports ($r_i^ = 1$) the quality to influencer i only when products are high-quality, and it will lie with a probability of $s_i^* = \left(\frac{\bar{\omega}_i}{1-\bar{\omega}_i}\right) \left(\frac{1-\omega}{\omega}\right)$ when products are low-quality.*

Lemma 3 shows that the platform reveals some information about product quality to influencers, counter-intuitively, only when the product defective rate is high (or product quality is on average low). This is because private communication can improve the platform's profit only when the influencers have sufficiently low priors about product quality. By flagging some low-quality products as “low-quality,” the platform increases the chances of inducing influencers to participate when receiving a message of “high-quality.” To see this mathematically, we can calculate the expected participation rate based on the optimal communication rule given by Lemma 3 as follows:

$$\mathbf{E}(e_i^*) = \min \left\{ 1, \frac{1-\omega}{1-\bar{\omega}_i} \right\}. \quad (11)$$

When the product defective rate is high ($\omega > \bar{\omega}_i$), the expected participation rate is strictly higher than that in the benchmark model, which is zero.

Lemma 4. *For any given technology $\kappa \in [0, \frac{1}{2}]$, compared to the benchmark model, if a platform can communicate product quality privately to the influencers, sales from influencer*

marketing campaigns is (weakly) higher and both the platform and influencers obtain (weakly) higher utility.

Lemma 4 is a standard result from a Bayesian persuasion setup (Kamenica and Gentzkow, 2011), which suggests that the platform indeed has an incentive to garble information about product quality from the brand and communicate it to the influencers, because they will be better off doing so.

With private communication, the platform could alleviate the “curse” of technology improvement to some extent. However, is it always more desirable for the platform to adopt a better AI under such circumstances? The next proposition²³ suggests that, surprisingly, that is not always the case.

Proposition 4. (Impact of Private Communication on Technology Choice) *Let $\kappa_{no\ comm.}^*$ and $\kappa_{comm.}^*$ denote respectively the optimal AI technology in the benchmark model (without quality communication) and in this extension (with quality communication).*

- *When $\theta < \theta_1$, $\omega > \bar{\omega}_N(\kappa = 0)$, and $\eta > \underline{\eta}_1$, the platform does not adopt AI if there is no quality communication ($\kappa_{no\ comm.}^* = 0$), but it will adopt AI if the communication is possible ($\kappa_{comm.}^* > 0$);*
- *When $\theta < \theta_2$, $\bar{\omega}_G(\kappa = 0) < \omega < \bar{\omega}_N(\kappa = 0)$, and $\eta \leq \underline{\eta}_2$, the platform adopts AI if there is no quality communication ($\kappa_{no\ comm.}^* > 0$) but it will not adopt AI if the communication is possible ($\kappa_{comm.}^* = 0$);*
- *Otherwise, the platform adopts AI regardless of whether or not quality communication is possible, and will develop a weakly more accurate AI if the communication is possible ($\kappa_{comm.}^* \geq \kappa_{no\ comm.}^* > 0$).*

The expressions for θ_1 , θ_2 , $\underline{\eta}_1$, and $\underline{\eta}_2$ are provided in Appendix on page A14. While private communication improves the welfare of both influencers and the platform, Proposition 4 shows that it could also discourage the platform from developing a better AI that matches brands to influencers. This happens when influencers do not have a large proportion of loyal followers (i.e., $\theta < \theta_2$), which implies a market where the incentives of the platform and influencers are misaligned based on Proposition 1. On the one hand, private communication reduces the uncertainty faced by influencers and thus induces them to participate at least when receiving a signal of “high-quality.” On the other hand, however, the expected

²³For tractability, we focus on the case where the commission rate δ is exogenously given.

participation rate declines in the AI prediction accuracy (κ) which requires the platform to lie more often when the true product quality is low. To achieve a higher expected revenue, a platform has to balance between increasing sales when the influencers do participate and increasing the probability of their participation.

Moreover, the platform that adopted AI when there was no private communication would choose not to develop AI if communication is feasible, when there are in aggregate more users following general influencers than those following niche influencers. This is because under those conditions, general influencers will not endorse a product if they do not receive any private information about product quality from the platform, as is in the benchmark model. Since the decisions of general influencers are less sensitive to the improvement of AI, the platform benefits from developing a less-than-perfect AI where only the niche influencers are willing to participate in the marketing campaigns. With the possibility of private communication, however, the platform is able to induce even the general influencers to participate after receiving signals of “high-quality.” When AI prediction accuracy (κ) improves, a platform has to lie more often to ensure that the general influencers at least participate upon receiving a “high-quality” signal. As a result, the expected overall participation rate of influencers declines, which increases the opportunity cost of developing an AI. Therefore, when the sales induced by general influencers is the platform’s major source of revenue, it may choose to give up completely on its AI matching algorithm.

Our analysis in this extension provides two managerial insights to platform owners and policymakers. First, platform can extract more rent by acquiring product quality information from the brands and partially revealing it to some influencers. By doing so, the platform alleviates influencers’ quality concern, thereby incentivizing them to participate in on-the-platform marketing campaigns. Therefore, regulating products on influencers’ recommendation channels is more than a legal issue and should be the policy consideration for regulating influencer marketing. Second, the platform may “leave money on the table” if it does not simultaneously adjust its AI technology when revealing product information to the influencers. This effect could also lead to an unexpected consequence: a platform may completely abandon the AI technology in some cases if it is allowed to privately communicate product quality to influencers.

5 Conclusion

Influencer marketing has become one of the hottest trending marketing strategies in recent years (Neate, 2020). A pain point for marketers interested in implementing influencer marketing is to find the “right” influencers to work with. One solution to this problem is for platforms to offer artificial intelligence (AI) to connect marketers with influencers.

We build a theoretical model where marketers are matched with influencers on a social media platform through AI technology. Since AI improves the matching between a marketer and an influencer, influencer marketing generates higher sales. On the one hand, higher sales benefits both influencers and the platform as they share a commission proportional to sales. On the other hand, if sales is high via an influencer’s recommendation channel, more users may stop following the influencer if they bought a low-quality product. The former sales effect induces the influencer to participate in a marketing campaign, whereas the latter quality concern effect deters one from doing so. The proportional size of an influencer’s loyal followers moderates the trade-off between these two effects. If an influencer has few loyal followers who always purchase and stick with her regardless of the realized product quality, the former effect dominates the latter. When she has many loyal followers, the latter effect dominates. However, if she has too many loyal followers, her main revenue comes from selling to loyal followers, thus making her endorsement decision insensitive to AI improvement.

We find that when the product defective rate is sufficiently high, a platform will refrain from adopting AI technology for matching influencers to marketers, even if the implementation cost is negligible. Furthermore, even when the product defective rate is not very high, a platform may prefer a “less-than-perfect” AI only if influencers on the platform do not have too many or too few loyal followers. We also find that the platform can alleviate influencers’ quality concern and thus extract more rent by gathering product quality information from the marketers and privately communicate it to influencers. However, the platform could “leave money on the table” if it does not adjust its AI technology accordingly when revealing product quality information to influencers. Interestingly, such quality communication may even prevent the platform from adopting AI as a consequence. Table 3 summarizes the managerial recommendations of the paper.

This paper serves to highlight the misalignment between a platform and its influencers when it comes to technology improvements. There are several ways to extend this line of research. First, one can explore a dynamic setting where influencers participate in recurring on-the-platform marketing and are allowed to choose their positioning as niche or general influencers in response to a platform’s AI strategies. Second, in our model, niche and general

Table 3: Summary of Managerial Recommendations

Main Takeaway:	Adopting or improving a marketer-influencer matching AI is not always in a platform’s best interest.
A platform should	not adopt AI if product defective rate is too high; not always pursue highest accuracy even if AI adoption is profitable; adjust AI strategy according to the size of influencers’ loyal followers; integrate AI strategy with pricing and communication strategies.

influencers have the same proportion of loyal followers on the platform. In reality, these two types of influencers may also differ in their proportional size of loyal followers. Future research can empirically estimate this parameter and incorporate it into the theoretical model. Finally, our model captures an important reason for platform users to purchase endorsed items: showing support for their favorite influencers. Future research can explore how other motives, such as influencers’ low-price-guarantee, may also drive users’ purchase.

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Appendix A: Proofs

In fact, Lemmas 1 and 2 and Propositions 1 and 2 are just a special case of the environment described in our first extension (Section 4.1), with $\tilde{\beta} = 0$. Therefore, we will prove a general version of Lemmas 1 and 2 and Propositions 1 and 2 under the environment in Section 4.1 with $\tilde{\beta} \geq 0$. Proposition 3 is naturally proven after we show the more general version of Propositions 1 and 2.

Proof of Lemma 1: Recall that an influencer i 's expected revenue function is given by

$$\mathbf{E} \pi_i(e_i) = \mathbf{E}_{q,m}[\pi_i^{on}(a_i; m_i) + \pi_i^{off}(a_i, q; m_i)].$$

In particular, if the influencer chooses to participate in the on-the-platform marketing campaign, her expected utility is

$$\mathbf{E} \pi_i(e_i = 1) = (1 - \delta)\alpha z_i(\theta + (1 - \theta) \mathbf{E}(m_i)) + (\beta - \tilde{\beta})z_i + \beta(z_i - \omega z_i(1 - \theta) \mathbf{E}(m_i)); \quad (\text{A1})$$

if she chooses not to participate, his expected utility is

$$\mathbf{E} \pi_i(e_i = 0) = 2\beta z_i. \quad (\text{A2})$$

Given the AI prediction accuracy (κ), we can write out the expected proportion of non-loyal followers who would purchase the product as follows:

$$\mathbf{E}(m_i) = \overline{m}_i\left(\frac{1}{2} + \kappa\right) + \underline{m}_i\left(\frac{1}{2} - \kappa\right) \quad (\text{A3})$$

Influencer i will participate in an on-the-marketing campaign iff $\mathbf{E} \pi_i(e_i = 1) \geq \mathbf{E} \pi_i(e_i = 0)$. Plugging the expressions in (A3) into (A1), we can solve for the participation condition with respect to ω . That is, influencer i will endorse the product ($e_i = 1$) if and only if $\omega \leq \overline{\omega}_i$, where

$$\overline{\omega}_i \equiv \frac{(1 - \delta)\alpha\theta - \tilde{\beta}}{\beta(1 - \theta) \left(\overline{m}_i \left(\frac{1}{2} + \kappa \right) + \underline{m}_i \left(\frac{1}{2} - \kappa \right) \right)} + \frac{(1 - \delta)\alpha}{\beta}.$$

□

Proof of Proposition 1: From Lemma 1, we know that for any given κ , influencer i will endorse the product if and only if the product defective rate is below a certain threshold,

i.e. $\omega < \bar{\omega}_i$. Hence we can rewrite the platform's profit-maximizing problem as follows:

$$\max_{\kappa \in [0, \frac{1}{2}]} \Pi = \sum_{i \in \{G, N\}} \mathbf{1}(\omega \leq \bar{\omega}_i) \delta \alpha \rho_i z_i [\theta + \mathbf{E}(m_i)(1 - \theta)],$$

where $\mathbf{1}(\cdot)$ is an indicator function that equals one if the statement in the parenthesis is true and zero otherwise. Note that the threshold value $\bar{\omega}_i$ increases in κ if $\theta \leq \frac{\tilde{\beta}}{\alpha(1-\delta)}$, while it declines in κ if otherwise. In other words, when θ is small, influencers are more likely to participate and endorse a product as the platform improves its AI. But influencers are less likely to do so when θ is large. Next we separately examine the platform's optimal AI choice in the following two cases (1) $\theta \leq \frac{\tilde{\beta}}{\alpha(1-\delta)}$ and (2) $\theta > \frac{\tilde{\beta}}{\alpha(1-\delta)}$ by backward induction.

Case 1: $\theta \leq \frac{\tilde{\beta}}{\alpha(1-\delta)}$

In this case, since $\bar{\omega}_i$ weakly increases in κ and $\mathbf{E}(m_i)$ strictly increases in κ for any influencer i , the platform's profit Π is strictly increasing in the prediction accuracy κ as long as *some* influencers participate in the marketing campaigns. For any given κ , this happens when the product defective rate is not too high, that is, when $\omega \leq \bar{\omega}_G(\kappa)$ or $\omega \leq \bar{\omega}_N(\kappa)$. Because $\bar{\omega}_i(\kappa)$ is a continuous and decreasing function of κ on $[0, \frac{1}{2}]$, the platform prefers a "perfect AI", i.e., $\kappa^* = \frac{1}{2}$, when $\omega \leq \max_{i \in \{G, N\}} \{\bar{\omega}_i(\kappa = \frac{1}{2})\} = \bar{\omega}_N(\kappa = \frac{1}{2})$.

When the product defective rate is sufficiently high ($\omega > \bar{\omega}_N(\kappa = \frac{1}{2})$), however, no influencers participate in the marketing campaigns, regardless of the prediction accuracy κ . In that case, the platform's profit is always zero, thus its optimal choice is not to develop AI, i.e., $\kappa^* = 0$.

Case 2: $\theta > \frac{\tilde{\beta}}{\alpha(1-\delta)}$

In this case, $\bar{\omega}_i$ strictly declines in κ whereas $\mathbf{E}(m_i)$ strictly increases in κ for any influencer i . By improving its AI prediction accuracy, the platform faces the trade-off between hurting the participation incentives of influencers and increasing sales from on-the-platform marketing campaigns.

Given that $\overline{m}_G < \overline{m}_N$ and $\frac{1}{2}(\overline{m}_G + \underline{m}_G) > \frac{1}{2}(\overline{m}_N + \underline{m}_N)$, we can derive the following condition for any κ :

$$\bar{\omega}_N(\kappa = \frac{1}{2}) < \bar{\omega}_G(\kappa = \frac{1}{2}) < \bar{\omega}_G(\kappa = 0) < \bar{\omega}_N(\kappa = 0).$$

(i) When $0 < \omega \leq \bar{\omega}_N(\kappa = \frac{1}{2})$, all influencers always participate in on-the-platform marketing

campaigns regardless of κ . Hence the platform's profit is given by

$$\Pi = \delta\alpha \sum_i \rho_i z_i [\theta + \mathbb{E}(m_i)(1 - \theta)],$$

which is strictly increasing in κ . Clearly, the optimal choice for the platform in this case is to develop a perfect AI, i.e. $\kappa^* = \frac{1}{2}$.

(ii) When $\bar{\omega}_N(\kappa = \frac{1}{2}) < \omega < \bar{\omega}_G(\kappa = \frac{1}{2})$, the general influencers always participate for any $\kappa \in [0, \frac{1}{2}]$, while the niche influencers will participate iff $\kappa \leq \kappa_N$, where

$$\kappa_N \equiv \frac{(1 - \delta)\alpha\theta - \tilde{\beta}}{(1 - \theta)(\beta\omega - (1 - \delta)\alpha)(\bar{m}_N - \underline{m}_N)} - \frac{1}{2} \left(\frac{\bar{m}_N + \underline{m}_N}{\bar{m}_N - \underline{m}_N} \right).$$

It is easy to see that, given the constraint on ω , we have $\kappa_N \in (0, \frac{1}{2})$. Hence we can rewrite the platform's profit function as follows:

$$\Pi = \begin{cases} \delta\alpha\rho_N z_N [\theta + \mathbb{E}(m_N)(1 - \theta)] + \delta\alpha\rho_G z_G [\theta + \mathbb{E}(m_G)(1 - \theta)] & \kappa \leq \kappa_N \\ \delta\alpha\rho_G z_G [\theta + \mathbb{E}(m_G)(1 - \theta)] & \kappa > \kappa_N \end{cases}.$$

The profit function Π increases in $\kappa \in [0, \kappa_N] \cup (\kappa_N, \frac{1}{2}]$, but it has a discrete fall at $\kappa = \kappa_N$. Hence the optimal κ^* is either κ_N or $\frac{1}{2}$, depending on whether or not the following condition satisfies:

$$\Pi(\kappa = \kappa_N) > \Pi(\kappa = \frac{1}{2}). \quad (\text{A4})$$

With some simple algebra, we know that inequality (A4) is equivalent to

$$\eta > \bar{\eta}(\theta, \omega) \equiv \left(\frac{\bar{m}_G - \underline{m}_G}{\bar{m}_N - \underline{m}_N} \right) \frac{\tilde{\beta} - \theta(1 - \delta)\alpha + (1 - \theta)(\beta\omega - (1 - \delta)\alpha)\bar{m}_N}{\theta\beta\omega - \tilde{\beta}},$$

where $\eta \equiv \frac{\rho_N z_N}{\rho_G z_G}$. Hence $\kappa^* = \kappa_N$ if $\eta > \bar{\eta}(\omega, \theta)$ and $\kappa^* = \frac{1}{2}$ if otherwise.

(iii) When $\bar{\omega}_G(\kappa = \frac{1}{2}) < \omega < \bar{\omega}_G(\kappa = 0)$, general influencers will participate iff $\kappa \leq \kappa_G$ and niche influencers will participate iff $\kappa \leq \kappa_N$, where

$$\kappa_i \equiv \frac{(1 - \delta)\alpha\theta - \tilde{\beta}}{(1 - \theta)(\beta\omega - (1 - \delta)\alpha)(\bar{m}_i - \underline{m}_i)} - \frac{1}{2} \left(\frac{\bar{m}_i + \underline{m}_i}{\bar{m}_i - \underline{m}_i} \right), i \in \{G, N\}.$$

Here it is easy to show that $\kappa_G \in (0, \frac{1}{2})$ and $\kappa_N \in (0, \frac{1}{2})$. Hence we can rewrite the platform's profit function as follows:

$$\Pi = \begin{cases} \sum_i \delta \alpha \rho_i z_i [\theta + \mathbb{E}(m_N)(1 - \theta)] & \kappa \leq \min\{\kappa_N, \kappa_G\} \\ \delta \alpha \rho_G z_G [\theta + \mathbb{E}(m_G)(1 - \theta)] & \kappa > \kappa_N, \kappa \leq \kappa_G \\ \delta \alpha \rho_N z_N [\theta + \mathbb{E}(m_N)(1 - \theta)] & \kappa \leq \kappa_N, \kappa > \kappa_G \\ 0 & \kappa > \max\{\kappa_N, \kappa_G\} \end{cases}.$$

The optimal κ^* is thus given by

$$\kappa^* = \operatorname{argmax}_{\kappa \in \{\kappa_N, \kappa_G\}} \{\Pi(\kappa_N), \Pi(\kappa_G)\}.$$

Clearly $\kappa^* < \frac{1}{2}$, that is, the platform prefers to develop a “less-than-perfect” AI in this case.

(iv) When $\bar{\omega}_G(\kappa = 0) < \omega < \bar{\omega}_N(\kappa = 0)$, general influencers never participate whereas niche influencers participate iff $\kappa \leq \kappa_N$. Again, it is easy to see that $\kappa_N \in (0, \frac{1}{2})$. Hence the platform’s profit is given by

$$\Pi = \begin{cases} \delta \alpha \rho_N z_N [\theta + \mathbb{E}(m_N)(1 - \theta)] & \kappa \leq \kappa_N \\ 0 & \kappa > \kappa_N \end{cases}.$$

Since the profit of the platform increases in $\kappa \in [0, \kappa_N]$ then decreases to and remains at zero when $\kappa \in (\kappa_N, \frac{1}{2}]$, the platform’s optimal choice is $\kappa^* = \kappa_N$.

(v) When $\omega > \bar{\omega}_N(\kappa = 0)$, no influencers would ever participate regardless of the AI prediction accuracy κ . Thus the platform earns zero profit from influencer marketing campaigns and chooses not to develop AI in this case, i.e., $\kappa^* = 0$.

Given the results in Case 1 and Case 2, we can summarize the equilibrium results as follows: the platform chooses not to develop AI such that $\kappa^* = 0$ iff $\omega > \max\{\bar{\omega}_N(\kappa = 0), \bar{\omega}_N(\kappa = \frac{1}{2})\}$; it chooses to develop a “less-than-perfect” AI such that $0 < \kappa^* < \frac{1}{2}$ iff²⁴

$$\underline{\theta} < \theta < \theta_1, \bar{\omega}_G(\kappa = \frac{1}{2}) < \omega < \bar{\omega}_N(\kappa = 0)$$

or

$$\theta_1 < \theta < \bar{\theta}, \bar{\omega}_N(\kappa = \frac{1}{2}) < \omega \leq \bar{\omega}_G(\kappa = \frac{1}{2}), \eta > \bar{\eta}(\theta, \omega),$$

²⁴Note that $\bar{\omega}_N(\kappa = \frac{1}{2}) < 1$ implies that $\theta < \bar{\theta}$ and $\bar{\omega}_G(\kappa = \frac{1}{2}) < 1$ implies that $\theta > \theta_1$.

where

$$\begin{aligned}\underline{\theta} &= \frac{\tilde{\beta}}{(1-\delta)\alpha}, \\ \theta_1 &= 1 - \frac{(1-\delta)\alpha - \tilde{\beta}}{(\beta - (1-\delta)\alpha)\overline{m}_G + (1-\delta)\alpha} \\ \overline{\theta} &= 1 - \frac{(1-\delta)\alpha - \tilde{\beta}}{(\beta - (1-\delta)\alpha)\overline{m}_N + (1-\delta)\alpha}.\end{aligned}$$

Otherwise, the platform chooses to develop a “perfect” AI such that $\kappa^* = \frac{1}{2}$. $\square$

Proof of Corollary 1: We prove this corollary by listing out the conditions such that only niche influencers participate in on-the-platform marketing while general influencers do not.

Note that the only case where it is possible that only niche influencers participate, is when niche influencers are less conservative than general ones, i.e., when $\omega > \overline{\omega}_G(\kappa = \hat{\kappa})$.

(1) When $\omega > \overline{\omega}_G(\kappa = 0)$ (which is even larger than $\overline{\omega}_G(\kappa = \hat{\kappa})$), general influencers would never participate in on-the-platform marketing regardless of the AI prediction accuracy. In equilibrium, general influencers will also not participate.

(2) When $\overline{\omega}_G(\kappa = \hat{\kappa}) < \omega \leq \overline{\omega}_G(\kappa = 0)$, the platform can either set a lower κ ($= \kappa_G$, as shown soon) such that both types of influencers participate in on-the-platform marketing, or set a higher κ ($= \kappa_N$, as shown soon) such that only niche influencers participate. To make sure that niche influencers are the only ones who participate, it is also required that the platform is more profitable when having a technology level such that only niche influencers participate than having one such that all influencers participate. As shown in the proof of Proposition 1, $\kappa^* \in \{\kappa_G, \kappa_N\}$, where

$$\kappa_i \equiv \frac{(1-\delta)\alpha\theta}{(1-\theta)(\beta\omega - (1-\delta)\alpha)(\overline{m}_i - \underline{m}_i)} - \frac{1}{2} \left(\frac{\overline{m}_i + \underline{m}_i}{\overline{m}_i - \underline{m}_i} \right), i \in \{G, N\},$$

and $\kappa_G < \kappa_N < \hat{\kappa}$. All influencers participate if $\kappa = \kappa_G$, while only niche influencers participate if $\kappa = \kappa_N$.

Denote $\xi(i, \kappa) \equiv \theta + (1-\theta)(\overline{m}_i(\frac{1}{2} + \kappa) + \underline{m}_i(\frac{1}{2} - \kappa))$, which is clearly positive and increasing in κ . Then

$$\Pi = \begin{cases} \delta\alpha\rho_N z_N \xi(N, \kappa_N) & \text{if } \kappa = \kappa_N \\ \delta\alpha\rho_N z_N \xi(N, \kappa_G) + \delta\alpha\rho_G z_G \xi(G, \kappa_G) & \text{if } \kappa = \kappa_G \end{cases}.$$

With some algebra, one can also show that $\xi(G, \kappa_G) = \xi(N, \kappa_N) = \theta + \frac{(1-\delta)\alpha\theta}{\beta\omega - (1-\delta)\alpha}$.

The platform earns a higher profit when only niche influencers participate if

$$\delta\alpha\rho_N z_N \xi(N, \kappa_N) > \delta\alpha\rho_N z_N \xi(N, \kappa_G) + \delta\alpha\rho_G z_G \xi(G, \kappa_G),$$

which is equivalent to

$$\eta \equiv \frac{\rho_N z_N}{\rho_G z_G} > \frac{1}{1 - \frac{\xi(N, \kappa_G)}{\xi(G, \kappa_G)}}.$$

Let

$$\bar{\eta}_1(\theta, \omega) \equiv \frac{1}{1 - \frac{\xi(N, \kappa_G)}{\xi(G, \kappa_G)}}.$$

Note that

$$\xi(G, \kappa) - \xi(N, \kappa) = (1 - \theta) \left[\frac{1}{2} ((\overline{m}_G + \underline{m}_G) - (\overline{m}_N + \underline{m}_N)) - \kappa ((\overline{m}_N - \underline{m}_N) - (\overline{m}_G - \underline{m}_G)) \right]$$

is decreasing in κ . Clearly, when $\kappa = \hat{\kappa} \equiv \frac{(\overline{m}_G + \underline{m}_G) - (\overline{m}_N + \underline{m}_N)}{2[(\overline{m}_N - \underline{m}_N) - (\overline{m}_G - \underline{m}_G)]}$, $\xi(G, \hat{\kappa}) - \xi(N, \hat{\kappa}) = 0$, so when $\kappa = \kappa_G < \hat{\kappa}$, we know that $\xi(G, \kappa_G) - \xi(N, \kappa_G) > 0$, i.e.,

$$\xi(G, \kappa_G) > \xi(N, \kappa_G).$$

Therefore, $\bar{\eta}_1(\theta, \omega) \equiv \frac{1}{1 - \frac{\xi(N, \kappa_G)}{\xi(G, \kappa_G)}} > 1$.

□

Proof of Lemma 2: An influencer's profit is clearly decreasing in δ , so there is a threshold of δ below which the influencer will participate in on-the-platform marketing. The threshold of δ for influencer i ($i = G, N$), denoted as $\hat{\delta}_i$, solves $\mathbf{E} \pi_i(e_i = 1) = \mathbf{E} \pi_i(e_i = 0)$ or $(1 - \delta)\alpha z_i(\theta + (1 - \theta)(\overline{m}_i(\frac{1}{2} + \kappa) + \underline{m}_i(\frac{1}{2} - \kappa))) + (\beta - \tilde{\beta})z_i + \beta(z_i - \omega z(1 - \theta)(\overline{m}_i(\frac{1}{2} + \kappa) + \underline{m}_i(\frac{1}{2} - \kappa))) = 2\beta z_i$. This gives

$$\hat{\delta}_i(\kappa) = 1 - \frac{\tilde{\beta} + \beta\omega(1 - \theta)(\overline{m}_i(\frac{1}{2} + \kappa) + \underline{m}_i(\frac{1}{2} - \kappa))}{\alpha(\theta + (1 - \theta)(\overline{m}_i(\frac{1}{2} + \kappa) + \underline{m}_i(\frac{1}{2} - \kappa)))}.$$

On the other hand, the platform's profit is increasing in δ , given that influencers do participate. Therefore, the platform's optimal choice of δ should let one of the influencers indifferent between participating and not participating, or set $\delta = \hat{\delta}_i$. Note that there is a possibility that even with a $\delta = 0$, influencer i still does not participate. This happens when $\hat{\delta}_i < 0$. Therefore, the platform's optimal choice of service fee percentage, δ^* , should

be either δ_G or δ_N , where $\delta_i = \max\{0, \hat{\delta}_i\}$.

□

Proof of Proposition 2: First, we look at how $\hat{\delta}_i(\kappa)$ changes with respect to κ . Since $(1 - \theta) (\overline{m}_i (\frac{1}{2} + \kappa) + \underline{m}_i (\frac{1}{2} - \kappa))$ is clearly increasing in κ , we know that $\hat{\delta}_i(\kappa)$ is increasing in κ if and only if $\zeta(x)$ is increasing in x , where $\zeta(x) \equiv 1 - \frac{\tilde{\beta} + \beta\omega x}{\alpha(\theta + x)}$. Because $\zeta'(x) = -\frac{\beta\omega\alpha(\theta + x) - (\tilde{\beta} + \beta\omega x)\alpha}{(\alpha(\theta + x))^2} = \frac{\tilde{\beta} - \beta\omega\theta}{\alpha(\theta + x)^2}$, we know that $\hat{\delta}_i(\kappa)$ is increasing (decreasing) in κ if $\tilde{\beta} - \beta\omega\theta$ is positive (negative).

If $\tilde{\beta} - \beta\omega\theta > 0$, i.e., $\omega < \frac{\tilde{\beta}}{\beta\theta}$, then $\hat{\delta}_i(\kappa)$ and thus the platform's profit are increasing in κ . Therefore, the platform will never have an incentive to develop a less-than-perfect AI. Thus, we only focus on the case when

$$\omega \geq \frac{\tilde{\beta}}{\beta\theta} \equiv \omega_1(\theta), \quad (\text{A5})$$

and thus $\hat{\delta}_i(\kappa)$ is decreasing in κ . It is easy to show that $\hat{\delta}_G(\kappa) \leq \hat{\delta}_N(\kappa)$ if $\kappa \leq \hat{\kappa}$. We focus on the case when $\max \hat{\delta} = \hat{\delta}_N(0) > 0$, which is equivalent to

$$\omega \leq \frac{\alpha\theta - \tilde{\beta}}{\frac{1}{2}(\overline{m}_N + \underline{m}_N)\beta(1 - \theta)} + \frac{\alpha}{\beta} \equiv \omega_2(\theta), \quad (\text{A6})$$

since otherwise it is the trivial case the platform earns 0 profit and does not develop AI ($\kappa^* = 0$). Note that $\omega_1(\theta)$ is decreasing in θ and $\omega_2(\theta)$ is increasing in θ . Also, note that both conditions (A5) and (A6) bind when $\theta = \frac{\tilde{\beta}}{\alpha}$ and $\omega = \frac{\alpha}{\beta}$. Therefore, these two conditions can simultaneously hold only when $\theta \geq \frac{\tilde{\beta}}{\alpha}$.

For any given κ , the candidates for optimal δ^* and the associated profit are then described as follows:

- If $\kappa < \hat{\kappa}$,
 - Either $\delta^* = \hat{\delta}_G(\kappa)$ so all influencers participate, and the platform's profit is thus $\Pi_1(\kappa) = \hat{\delta}_G(\kappa)\alpha[\rho_N z_N \xi(N, \kappa) + \rho_G z_G \xi(G, \kappa)]$,
 - Or $\delta^* = \hat{\delta}_N(\kappa)$ so only niche influencers participate, and the platform's profit is thus $\Pi_2(\kappa) = \hat{\delta}_N(\kappa)\alpha\rho_N z_N \xi(N, \kappa)$.
- If $\kappa > \hat{\kappa}$,
 - Either $\delta^* = \hat{\delta}_N(\kappa)$ so all influencers participate, and the platform's profit is thus $\Pi_3(\kappa) = \hat{\delta}_N(\kappa)\alpha[\rho_N z_N \xi(N, \kappa) + \rho_G z_G \xi(G, \kappa)]$,

- Or $\delta^* = \hat{\delta}_G(\kappa)$ so only general influencers participate, and the platform's profit is thus $\Pi_4(\kappa) = \hat{\delta}_G(\kappa)\alpha\rho_G z_G \xi(G, \kappa)$.

Therefore, for any given κ , the platform's profit is

$$\Pi(\kappa) = \begin{cases} \max\{\Pi_1(\kappa), \Pi_2(\kappa)\} & \text{if } \kappa < \hat{\kappa} \\ \max\{\Pi_3(\kappa), \Pi_4(\kappa)\} & \text{if } \kappa \geq \hat{\kappa} \end{cases},$$

and then the platform chooses κ to maximize $\Pi(\kappa)$.

Note that a sufficient condition for the optimal technology to be “less-than-perfect” is that we simply choose $\kappa = \hat{\kappa} \in (0, \frac{1}{2})$ and the profit it induces is higher than that induced by $\kappa = 0$ (no AI) and $\kappa = \frac{1}{2}$ (perfect AI), i.e., $\Pi(\hat{\kappa}) > \Pi(0)$ and $\Pi(\hat{\kappa}) > \Pi(\frac{1}{2})$.

By construction, we know that $\Pi_1(\hat{\kappa}) > \Pi_2(\hat{\kappa})$, $\Pi_3(\hat{\kappa}) > \Pi_4(\hat{\kappa})$, and $\Pi_1(\hat{\kappa}) = \Pi_3(\hat{\kappa})$. Therefore, that sufficient condition is equivalent to

$$\Pi_1(\hat{\kappa}) > \max\{\Pi_1(0), \Pi_2(0)\}, \quad (\text{A7})$$

and

$$\Pi_3(\hat{\kappa}) > \max\{\Pi_3(\frac{1}{2}), \Pi_4(\frac{1}{2})\}. \quad (\text{A8})$$

With some algebra, one can show that condition (A7) is equivalent to $\omega < \min\{\omega_{31}(\theta), \omega_{32}(\theta)\}$; condition (A8) is equivalent to $\omega > \max\{\omega_{41}(\theta), \omega_{42}(\theta)\}$ if $\eta < \frac{(\overline{m}_G - \underline{m}_G)(\overline{m}_N - \underline{m}_G)}{\underline{m}_G \overline{m}_N - \overline{m}_G \underline{m}_N}$, and equivalent to $\omega > \omega_{41}(\theta)$ otherwise, where

$$\begin{aligned} \omega_{31}(\theta) &= \frac{\alpha A[(1-\theta)M_G + 2\theta] - 2\tilde{\beta}\eta B}{\beta[2\theta(1+\eta)\Delta m_G + (1-\theta)M_G A]}, \\ \omega_{32}(\theta) &= \frac{\alpha[(1-\theta)\eta\Delta m_N\Delta m_G + 2(1-\theta)C] - 2\tilde{\beta}B}{\beta(1-\theta)[\eta\Delta m_N(M_G - M_N) + 2C]}, \\ \omega_{41}(\theta) &= \frac{\alpha A[(1-\theta)\overline{m}_N + \theta] + \tilde{\beta}B}{\beta[\theta(1+\eta)\Delta m_N + (1-\theta)\overline{m}_N A]}, \\ \omega_{42}(\theta) &= \frac{\alpha[(1-\theta)\Delta m_G(\overline{m}_N - \underline{m}_G) - \eta((1-\theta)C + \theta A)] + \tilde{\beta}\eta B}{\beta(1-\theta)[\Delta m_G(\overline{m}_N - \underline{m}_G) - \eta C]}, \end{aligned}$$

with $M_i \equiv \overline{m}_i + \underline{m}_i$ ($i = G, N$), $\Delta m_i \equiv \overline{m}_i - \underline{m}_i$ ($i = G, N$), $A \equiv \Delta m_G + \eta\Delta m_N$, $B \equiv \Delta m_N - \Delta m_G$, and $C \equiv \underline{m}_G \overline{m}_N - \underline{m}_N \overline{m}_G$.

Define $\overline{\omega}(\theta) = \min\{\omega_2(\theta), \omega_{31}(\theta), \omega_{32}(\theta)\}$ and $\underline{\omega}(\theta) = \max\{\omega_1(\theta), \omega_{41}(\theta), \omega_{42}(\theta)\}$, then $\omega > \underline{\omega}(\theta)$ and $\omega < \overline{\omega}(\theta)$ guarantee $\kappa^* \in (0, \frac{1}{2})$.

Further calculations of the first order derivatives show that ω_{31} and ω_{32} are increasing

in θ while ω_{41} and ω_{42} are decreasing in θ . Therefore $\underline{\omega}(\theta)$ are decreasing in θ and $\bar{\omega}(\theta)$ are increasing in θ . Also, $\omega_1(\theta) = \omega_2(\theta) = \omega_{31}(\theta) = \omega_{32}(\theta) = \omega_{41}(\theta) = \omega_{42}(\theta) = \frac{\alpha}{\beta}$ when $\theta = \frac{\tilde{\beta}}{\alpha}$, so $\underline{\omega}(\theta) = \bar{\omega}(\theta)$ when $\theta = \frac{\tilde{\beta}}{\alpha}$. Therefore, for any $\theta > \frac{\tilde{\beta}}{\alpha}$, $\underline{\omega}(\theta) < \bar{\omega}(\theta)$ we can find a region for $\omega \in (\underline{\omega}(\theta), \bar{\omega}(\theta))$ where $\kappa^* \in (0, \frac{1}{2})$, and the measure for this region is increasing in θ . Thus, we finish the proof. $\square$

Proof of Lemma 3: To design an optimal information environment, the platform maximizes the expected probability of influencer i participating in a campaign when the message indicates good quality ($\hat{q} = H$). Formally, the problem of the platform is given by

$$\max_{r,s} \mathbf{E}(e_i) = \mathbf{E}(\mathbf{E}(e_i|\hat{q}(q))) \quad (\text{A9})$$

$$\text{s.t. } \mathbf{P}_i(q = L|\hat{q} = H) \leq \bar{\omega}_i \quad (\text{A10})$$

$$\mathbf{P}_i(q = L|\hat{q} = H) \geq \bar{\omega}_i \quad (\text{A11})$$

By law of iterated expectation, the expected participation probability of influencer i can be written as follows:

$$\begin{aligned} \mathbf{E}(e_i) &= \mathbf{E}(\mathbf{E}(e_i|\hat{q}(q))) \\ &= \mathbf{E}(e_i|\hat{q} = H)\mathbf{P}_i(\hat{q} = H) + \mathbf{E}(e_i|\hat{q} = L)\mathbf{P}_i(\hat{q} = L) \\ &= \mathbf{P}_i(\hat{q} = H) \\ &= r_i(1 - \omega) + s_i\omega \end{aligned} \quad (\text{A12})$$

Plugging (A12) into (A9), we can solve for the optimal communication rule:

$$r_i^* = 1; s_i^* = \min \left\{ 1, \left(\frac{\bar{\omega}_i}{1 - \bar{\omega}_i} \right) \left(\frac{1 - \omega}{\omega} \right) \right\}.$$

Note that when the average product quality is high enough ($\omega \leq \bar{\omega}_i$), the platform's best response is to always send a message $\hat{q} = H$ to the influencer, regardless of the true quality q . This message is effectively uninformative, and hence it does not affect the decision of an influencer. In those cases, we consider the platform not revealing product quality to the influencer. $\square$

Proof of Proposition 4: Before receiving any message (signal) from the platform, an influencer i 's prior belief about the product quality is given by the product distribution in the market, i.e.,

$$\mathbf{P}_i(q = L|\emptyset) = \omega.$$

That is, $q = L$ and $q = H$ with prob. ω and $1 - \omega$ respectively. Upon receiving the platform's message $\hat{q}$, influencer i updates her belief about the product quality according to Bayes' rules. She will endorse a product if and only if her posterior belief about this product being low-quality is below a threshold, i.e.

$$\mathbf{P}_i(q = L|\hat{q}) \leq \bar{\omega}_i.$$

From lemma 3, we know that for any given κ , the platform sends informative messages to influencer i that reveals product quality only when the product quality is on average poor in the market ($\omega > \bar{\omega}_i$). Based on influencer i 's expected participation rate characterized by equation (11), we can rewrite the platform's profit-maximizing problem as follows:

$$\begin{aligned} \max_{\kappa \in [0, \frac{1}{2}]} \Pi &= \sum_{i \in \{G, N\}} \delta \alpha \mathbf{E}(e_i^*) \rho_i z_i [\theta + \mathbf{E}(m_i) (1 - \theta)] \\ &= \sum_{i \in \{G, N\}} \delta \alpha \min \left\{ 1, \frac{1 - \omega}{1 - \omega_i} \right\} \rho_i z_i [\theta + \mathbf{E}(m_i) (1 - \theta)], \end{aligned}$$

where $\bar{\omega}_i = \frac{(1-\delta)\alpha\theta - \tilde{\beta}}{\beta(1-\theta)(\bar{m}_i(\frac{1}{2} + \kappa) + m_i(\frac{1}{2} - \kappa))} + \frac{(1-\delta)\alpha}{\beta}.$

Since influencers react differently to AI improvement when θ is small vs. large, similar to the proof of 1, we separately examine platform's AI choice in the following two cases (1) $\theta \leq \frac{\tilde{\beta}}{\alpha(1-\delta)}$ and (2) $\theta > \frac{\tilde{\beta}}{\alpha(1-\delta)}$ by backward induction.

Case 1: $\theta \leq \frac{\tilde{\beta}}{\alpha(1-\delta)}$

In this case, $\bar{\omega}_i$ weakly increases in κ , and thus we know that influencer i 's participation rate $\mathbf{E}(e_i)$ also increases in κ . In the benchmark model, when the product defective rate is sufficiently high ($\omega > \bar{\omega}_N(\kappa = \frac{1}{2})$), no influencers participate in the marketing campaigns and thus the platform optimally chooses not to develop AI. However, now that the platform has the option of sending an informative message to influencers, their participation rates are always above zero (i.e. $\mathbf{E}(e_i) > 0$). In other words, the platform's profit is strictly increasing in κ even when product quality is on average poor. As a result, the platform always prefers a perfect AI, i.e. $\kappa^* = \frac{1}{2}$, independent of the product quality in this case.

Case 2: $\theta > \frac{\tilde{\beta}}{\alpha(1-\delta)}$

In this case, $\bar{\omega}_i$ strictly declines in κ , and thus we know that influencer i 's participation rate $\mathbf{E}(e_i)$ also declines in κ when $\omega > \bar{\omega}_i$. By assumption, we know that

$$\bar{\omega}_N(\kappa = \frac{1}{2}) < \bar{\omega}_G(\kappa = \frac{1}{2}) < \bar{\omega}_G(\kappa = 0) < \bar{\omega}_N(\kappa = 0).$$

(i) When $0 < \omega \leq \bar{\omega}_N(\kappa = \frac{1}{2})$, the platform does not reveal any product information, but all influencers still participate in on-the-platform marketing campaigns for any κ . Hence the platform's profit is given by

$$\Pi = \delta\alpha \sum_i \rho_i z_i [\theta + \mathbb{E}(m_i)(1 - \theta)],$$

which is strictly increasing in κ . The optimal choice for the platform is to develop a perfect AI, i.e. $\kappa^* = \frac{1}{2}$.

(ii) When $\bar{\omega}_N(\kappa = \frac{1}{2}) < \omega \leq \bar{\omega}_G(\kappa = \frac{1}{2})$, the platform will send an informative message $\hat{q}$ to the niche influencers if product quality is on average poor, but it does not reveal any information to the general influencers (since the latter always participate for any κ). Hence the platform's profit function is given by

$$\Pi = \begin{cases} \delta\alpha\rho_N z_N [\theta + \mathbb{E}(m_N)(1 - \theta)] + \delta\alpha\rho_G z_G [\theta + \mathbb{E}(m_G)(1 - \theta)] & \kappa \leq \kappa_N \\ \delta\alpha\rho_N z_N \left(\frac{1-\omega}{1-\bar{\omega}_N}\right) [\theta + \mathbb{E}(m_N)(1 - \theta)] + \delta\alpha\rho_G z_G [\theta + \mathbb{E}(m_G)(1 - \theta)] & \kappa > \kappa_N \end{cases},$$

where $\kappa_N \equiv \frac{(1-\delta)\alpha\theta - \tilde{\beta}}{(1-\theta)(\beta\omega - (1-\delta)\alpha)(\bar{m}_N - m_N)} - \frac{1}{2} \left(\frac{\bar{m}_N + m_N}{\bar{m}_N - m_N} \right)$. Note that $\Pi(\cdot)$ strictly increases in $\kappa \in [0, \kappa_N]$. We can also show that $\Pi(\cdot)$ is a convex function of κ when $\kappa \in (\kappa_N, \frac{1}{2}]$. This is because

$$\frac{\partial^2 \Pi(\kappa > \kappa_N)}{\partial \kappa^2} = \frac{2\beta(\theta - 1)^2(\bar{m}_N - m_N)^2((1 - \delta)\alpha\theta - \tilde{\beta})(\theta(\beta - (1 - \delta)\alpha) + (1 - \delta)\alpha\theta - \tilde{\beta})}{(-(1 - \delta)\alpha\theta + \tilde{\beta} + (1 - \theta)\mathbb{E}(m_N)(\beta - (1 - \delta)\alpha))^3} > 0.$$

Therefore, if the first-order derivative of Π is positive at $\kappa = \kappa_N$ (i.e. $\frac{\partial \Pi(\kappa = \kappa_N)}{\partial \kappa} \geq 0$), we have Π monotonically increases in $\kappa \in (\kappa_N, \frac{1}{2}]$; if the first-order derivative of Π is negative at $\kappa = \kappa_N$ (i.e. $\frac{\partial \Pi(\kappa = \kappa_N)}{\partial \kappa} < 0$), we have Π either first decreases then increases, or monotonically decreases in $\kappa \in (\kappa_N, \frac{1}{2}]$. Hence the optimal AI choice is given by

$$\kappa^* = \begin{cases} \kappa_N & \Pi(\kappa_N) \geq \Pi\left(\frac{1}{2}\right) \\ \frac{1}{2} & \text{o/w} \end{cases}.$$

Recall that

$$\Pi(\kappa_N) = \delta\alpha \sum_i \rho_i z_i \left[\theta + \left[\overline{m}_i \left(\frac{1}{2} + \kappa_N \right) + \underline{m}_i \left(\frac{1}{2} - \kappa_N \right) \right] (1 - \theta) \right],$$

and

$$\Pi\left(\frac{1}{2}\right) = \delta\alpha \rho_N z_N \left(\frac{1 - \omega}{1 - \overline{\omega}_N(\frac{1}{2})} \right) [\theta + \overline{m}_N(1 - \theta)] + \delta\alpha \rho_G z_G [\theta + \overline{m}_G(1 - \theta)].$$

With some algebra, we can re-arrange $\Pi(\kappa_N) \geq \Pi(\frac{1}{2})$ as follows:

$$\frac{\rho_N z_N}{\rho_G z_G} \geq \frac{\left(\frac{1}{2} - \kappa_N\right) (\overline{m}_G - \underline{m}_G) (1 - \theta)}{\theta \left(\frac{\omega - \overline{\omega}_N(\frac{1}{2})}{1 - \overline{\omega}_N(\frac{1}{2})} \right) + \left(\overline{m}_N \left(\frac{1}{2} + \kappa_N - \frac{1 - \omega}{1 - \overline{\omega}_N(\frac{1}{2})} \right) + \underline{m}_N \left(\frac{1}{2} - \kappa_N \right) \right) (1 - \theta)} := \underline{\eta}(\theta, \omega).$$

(iii) When $\overline{\omega}_G(\kappa = \frac{1}{2}) < \omega \leq \overline{\omega}_G(\kappa = 0)$, the platform will send an informative message $\hat{q}_N$ to the niche influencers only if $\kappa > \kappa_N$ and $\hat{q}_G$ to general influencers only if $\kappa > \kappa_G$. Hence the platform's profit function is given by

$$\Pi = \begin{cases} \delta\alpha \sum_i \rho_i z_i [\theta + \mathbb{E}(m_i) (1 - \theta)] & \kappa \leq \min\{\kappa_G, \kappa_N\} \\ \delta\alpha \sum_i \rho_i z_i \min\left\{1, \frac{1 - \omega}{1 - \overline{\omega}_i}\right\} [\theta + \mathbb{E}(m_i) (1 - \theta)] & \min\{\kappa_G, \kappa_N\} < \kappa \leq \max\{\kappa_G, \kappa_N\} \\ \delta\alpha \sum_i \rho_i z_i \left(\frac{1 - \omega}{1 - \overline{\omega}_i} \right) [\theta + \mathbb{E}(m_i) (1 - \theta)] & \kappa > \max\{\kappa_G, \kappa_N\} \end{cases}.$$

Note that $\Pi(\cdot)$ strictly increases in κ when $\kappa \leq \min\{\kappa_G, \kappa_N\}$. Similarly, we can show that $\Pi(\cdot)$ is a convex function of κ when $\min\{\kappa_G, \kappa_N\} < \kappa \leq \max\{\kappa_G, \kappa_N\}$ and when $\kappa > \max\{\kappa_G, \kappa_N\}$. Therefore, the optimal AI choice κ^* is given by

$$\kappa^* = \begin{cases} \kappa_N & \Pi(\kappa_N) \geq \Pi(\kappa_G), \Pi(\kappa_N) \geq \Pi\left(\frac{1}{2}\right) \\ \kappa_G & \Pi(\kappa_G) > \Pi(\kappa_N), \Pi(\kappa_G) \geq \Pi\left(\frac{1}{2}\right) \\ \frac{1}{2} & \text{o/w} \end{cases}.$$

(iv) When $\overline{\omega}_G(\kappa = 0) < \omega \leq \overline{\omega}_N(\kappa = 0)$, the platform will send an informative signal to the niche influencers only when $\kappa > \kappa_N$, but it sends an informative signal to the general

influencers for any κ . Hence the platform's profit is given by

$$\Pi = \begin{cases} \delta\alpha\rho_N z_N [\theta + \mathbb{E}(m_N)(1-\theta)] + \delta\alpha\rho_G z_G \left(\frac{1-\omega}{1-\bar{\omega}_G}\right) [\theta + \mathbb{E}(m_G)(1-\theta)] & \kappa \leq \kappa_N \\ \delta\alpha \sum_i \rho_i z_i \left(\frac{1-\omega}{1-\bar{\omega}_i}\right) [\theta + \mathbb{E}(m_i)(1-\theta)] & \kappa > \kappa_N \end{cases}.$$

Note that $\Pi(\cdot)$ strictly increases in κ when $\kappa \leq \min\{\kappa_G, \kappa_N\}$. Similarly, we can show that $\Pi(\cdot)$ is a convex function of κ in $\kappa \in (\kappa_N, \frac{1}{2}]$. Therefore, the optimal AI choice κ^* is given by

$$\kappa^* = \begin{cases} 0 & \Pi(0) \geq \max\{\Pi(\kappa_N), \Pi(\frac{1}{2})\} \\ \kappa_N & \Pi(\kappa_N) > \Pi(0), (\kappa_N) \geq \Pi(\frac{1}{2}) \\ \frac{1}{2} & \text{o/w} \end{cases}.$$

(v) When $1 > \omega > \bar{\omega}_N(\kappa = 0)$, the platform will always send an informative signal to both the niche and the general influencers. Hence the profit of the platform is given by

$$\Pi = \delta\alpha \sum_i \rho_i z_i \left(\frac{1-\omega}{1-\bar{\omega}_i}\right) [\theta + \mathbb{E}(m_i)(1-\theta)].$$

Similarly, we can show that $\Pi(\cdot)$ is a convex function of κ . Hence the optimal AI choice κ^* is given by

$$\kappa^* = \begin{cases} 0 & \Pi(0) \geq \Pi(\frac{1}{2}) \\ \frac{1}{2} & \text{o/w} \end{cases}.$$

Finally, we compare side-by-side the results in Case 1 and Case 2 with those derived from Proposition 1. Let $\kappa_{\text{no comm.}}^*$ and $\kappa_{\text{comm.}}^*$ denote the optimal AI choice in the benchmark model and in this extension, respectively. We summarize the three possible scenarios below:

- $\kappa_{\text{no comm.}}^* = 0 < \kappa_{\text{comm.}}^*$ if

$$\theta \leq \underline{\theta}, \omega > \bar{\omega}_N\left(\kappa = \frac{1}{2}\right)$$

or

$$\underline{\theta} < \theta < \theta_1, \omega > \bar{\omega}_N(\kappa = 0), \eta > \underline{\eta}_1.$$

- $\kappa_{\text{no comm.}}^* > 0 = \kappa_{\text{comm.}}^*$ if

$$\underline{\theta} < \theta < \theta_2, \bar{\omega}_G(\kappa = 0) < \omega < \bar{\omega}_N(\kappa = 0), \eta \leq \underline{\eta}_2.$$

- $0 < \kappa_{\text{no comm.}}^* \leq \kappa_{\text{comm.}}^*$ if otherwise,

where

$$\begin{aligned}
\theta_1 &= \min \left\{ \frac{(\beta - (1 - \delta)\alpha) \frac{1}{2} (\overline{m}_N + \underline{m}_N) + \tilde{\beta}}{(1 - \delta)\alpha + (\beta - (1 - \delta)\alpha) \frac{1}{2} (\overline{m}_N + \underline{m}_N)}, \xi_N \right\} \\
\theta_2 &= \frac{(\beta - (1 - \delta)\alpha) \frac{1}{2} (\overline{m}_G + \underline{m}_G) + \tilde{\beta}}{(1 - \delta)\alpha + (\beta - (1 - \delta)\alpha) \frac{1}{2} (\overline{m}_G + \underline{m}_G)} \\
\underline{\eta}_1 &= \begin{cases} 0 & \text{if } 0 < \theta < \xi_G \\ \frac{-X_G}{X_N} & \text{if } \xi_G < \theta < \xi_N \end{cases} \\
\underline{\eta}_2 &= \begin{cases} \frac{(1-\omega)\beta(Y_0-Y_N)(1-\theta)(\beta\omega-(1-\delta)\alpha)}{(1-\delta)\alpha\theta-\tilde{\beta}-\frac{1}{2}(\overline{m}_N+\underline{m}_N)(1-\theta)(\beta\omega-(1-\delta)\alpha)} & \text{if } \bar{\xi} \leq \theta \leq 1 \\ \min \left\{ \frac{(1-\omega)Z_G}{Z_N}, \frac{(1-\omega)\beta(Y_0-Y_N)(1-\theta)(\beta\omega-(1-\delta)\alpha)}{(1-\delta)\alpha\theta-\tilde{\beta}-\frac{1}{2}(\overline{m}_N+\underline{m}_N)(1-\theta)(\beta\omega-(1-\delta)\alpha)} \right\} & \text{if } 0 < \theta < \bar{\xi} \end{cases}
\end{aligned}$$

and

$$\begin{aligned}
X_i &= \frac{\frac{1}{2} (\overline{m}_i + \underline{m}_i) [\theta + \frac{1}{2} (\overline{m}_i + \underline{m}_i) (1 - \theta)]}{(\beta - (1 - \delta)\alpha) (1 - \theta) \frac{1}{2} (\overline{m}_i + \underline{m}_i) - (1 - \delta)\alpha\theta + \tilde{\beta}} \\
&\quad - \frac{\overline{m}_i [\theta + \overline{m}_i (1 - \theta)]}{(\beta - (1 - \delta)\alpha) (1 - \theta) \overline{m}_i - (1 - \delta)\alpha\theta + \tilde{\beta}}, i = G, N \\
Y_0 &= \frac{(\theta + \frac{1}{2} (\overline{m}_G + \underline{m}_G) (1 - \theta)) \frac{1}{2} (\overline{m}_G + \underline{m}_G)}{(\beta - (1 - \delta)\alpha) (1 - \theta) \frac{1}{2} (\overline{m}_G + \underline{m}_G) - (1 - \delta)\alpha\theta + \tilde{\beta}} \\
Y_N &= \frac{\theta [\frac{1}{2} (\overline{m}_G + \underline{m}_G) + \kappa_N (\overline{m}_G - \underline{m}_G)] + [\frac{1}{2} (\overline{m}_G + \underline{m}_G) + \kappa_N (\overline{m}_G - \underline{m}_G)]^2 (1 - \theta)}{(\beta - (1 - \delta)\alpha) (1 - \theta) [\frac{1}{2} (\overline{m}_G + \underline{m}_G) + \kappa_N (\overline{m}_G - \underline{m}_G)] - (1 - \delta)\alpha\theta + \tilde{\beta}} \\
Z_N &= \frac{\theta \left((1 - \theta) \overline{m}_N (\beta\omega - (1 - \delta)\alpha) - \left((1 - \delta)\alpha\theta - \tilde{\beta} \right) \right) - \beta ((1 - \theta) \overline{m}_N)^2 (1 - \omega)}{(1 - \theta) \overline{m}_N (\beta - (1 - \delta)\alpha) - \left((1 - \delta)\alpha\theta - \tilde{\beta} \right)} \\
&\quad + (1 - \theta) \frac{1}{2} (\overline{m}_N + \underline{m}_N) \\
Z_G &= \frac{\beta(1 - \theta) \overline{m}_G (\theta + \overline{m}_G (1 - \theta))}{(1 - \theta) \overline{m}_G (\beta - (1 - \delta)\alpha) - \left((1 - \delta)\alpha\theta - \tilde{\beta} \right)} \\
&\quad - \frac{\beta(1 - \theta) \frac{1}{2} (\overline{m}_G + \underline{m}_G) (\theta + \frac{1}{2} (\overline{m}_G + \underline{m}_G) (1 - \theta))}{(1 - \theta) \frac{1}{2} (\overline{m}_G + \underline{m}_G) (\beta - (1 - \delta)\alpha) - \left((1 - \delta)\alpha\theta - \tilde{\beta} \right)} \\
\xi_i &\in \Phi(\overline{m}_i, \underline{m}_i) := [0, 1] \cap \{\xi | X_i(\theta = \xi) = 0\}, i = G, N \\
\bar{\xi} &\in \Omega(\omega, \overline{m}_N, \underline{m}_N) := [0, 1] \cap \{\xi | Z_N(\theta = \xi) = 0\}.
\end{aligned}$$

□

Appendix B: Empirical Analysis

In this section, we document in detail the data collection process and empirical analysis that we conducted to provide supporting evidence for model predictions described in Corollary 1. We begin by describing the data collection process which involves the following three steps:

Step 1: On February 1, 2021, we acquired data on a list of Top 1000 influencers (KOL) active on Douyin, the Chinese version of TikTok, from a leading company²⁵ specializing in real-time analytics on influencer marketing for Chinese e-commerce brands. In particular, we have access to each influencer’s “Douyin ID”, nickname, “Blue V” status²⁶, number of followers, number of likes, and total number of videos.

Step 2: Between February 2, 2021 and February 7, 2021, we recruited a team of 8 independent coders²⁷ who had prior experiences in using the social media app Douyin and whose native language is Mandarin Chinese. In the recruitment ads, we specified the classification tasks coders needed to complete. Before proceeding with the next step, we also held a virtual group meeting with the coders and provided them a detailed protocol (original transcript attached on page A20²⁸). We clarified the definition of “generality” to our coders by giving two extreme examples (one rated as 1 and another rated as 5) and stating that it referred to “how diversified you would expect the followers of the influencer’s channel to be.”

Step 3: Between February 8, 2021 and February 16, 2021, the coders were given a list of influencers on Douyin, and asked to complete the following four tasks based on the protocol:

1. Identify whether the influencer account is associated with any official (e.g. government agency, TV networks, etc.) or celebrity channels.
2. Identify whether the influencer has an on-the-platform product recommendation outlet, based on whether or not there exists an visible icon named “store window” on the profile page of the influencer’s channel.
3. Record the total number of recommended products in each influencer’s “store window” if there is one.
4. Rate the influencer’s generality on a scale of 1-5 based on the most recent 50 videos made by the influencers.

²⁵<https://toobigdata.com>.

²⁶If an influencer has a “Blue V” status, then the influencer’s channel is usually associated with the government (e.g. “People’s Daily”) or an official TV network (e.g. “CCTV”).

²⁷4 female and 4 male undergraduate and graduate students from Johns Hopkins University.

²⁸The original transcript was in Chinese because we required our RA to be fluent in Chinese in order to finish the tasks. We provide the English translation on page A21.

Specifically, we grouped the 8 coders into 4 groups. Each influencer was assigned to two independent coders. Note that, however, the coders did not know the identity of the other person in the same group, and nor did they see the data generated by the other person. We required that all the coders use the mobile app for Douyin on their smart phones when completing the tasks.²⁹ The entire process spanned for about a week during which we assigned around 50 influencers per day to each coder on 5 separate days to control for data quality. Since the whole process was completed within 9 days, we can reasonably assume that there is no temporal change in the data on influencers.

Next we explain our empirical analysis using the collected data discussed above. First, we preclude all influencer accounts associated with any official or celebrity channels, since they mainly use their Douyin account as a communication tool with their audience rather than profiting from the channel. Based on this selection criteria, we obtain a final dataset with 752 observations (influencers) for our analysis.

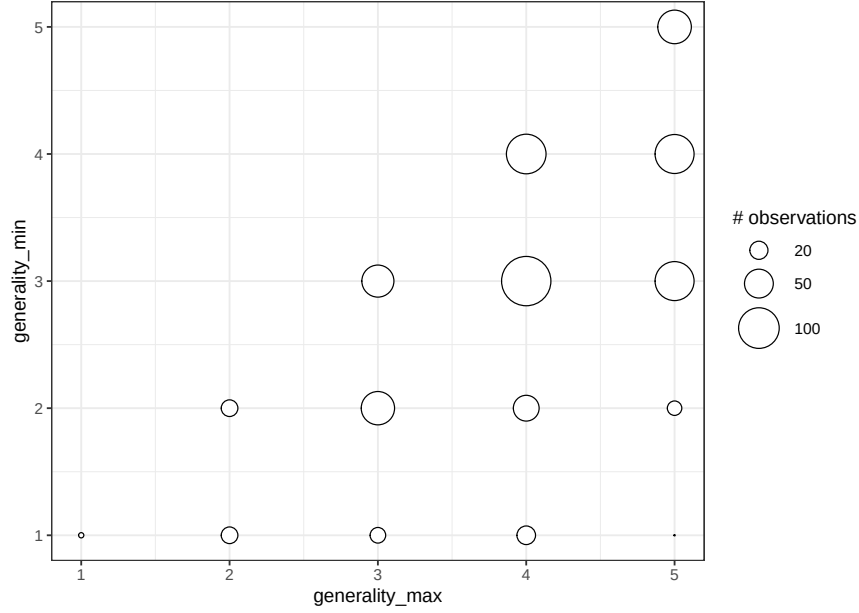
We document reasonably high inter-coder reliability when they rated the influencers’ generality: For each influencer, we define **generality_max** (**generality_min**) as the maximum (minimum) of the ratings on generality given by two different coders. The correlation between **generality_max** and **generality_min** in our dataset is 0.62, and the scatter plot between them is illustrated by Figure A1.

To identify the type (i.e., general/niche) of each influencer, we defined an influencer as “niche” if the average generality rating of the two coders is less than or equal to 3, and otherwise as “general,” and let dummy variable **generality** equal to 1 if the influencer is “general” and 0 if she is “niche.” As stated on page 16, we assume that the product defective rate is not so high that it becomes impossible for general influencers to participate in on-the-platform marketing under any AI environment, which precludes condition (1) in Corollary 1. Our empirical data shows that the ratio between the total number of followers for niche influencers and that for general influencers is $\eta = \frac{2.53 \text{ billion}}{5.96 \text{ billion}} = 0.42 < 1$, which violates condition (2.2) in Corollary 1.³⁰ Therefore, Corollary 1 predicts general influencers are more likely to participate in on-the-platform marketing. Below we describe how we test this hypothesis empirically using our data.

If an influencer participates in on-the-platform marketing, there will be a link to her “store window” on her homepage, which is a feature provided within the mobile app. By

²⁹Douyin only provides a mobile app and does not offer a PC interface.

³⁰An alternative classification scheme is to consider influencers with a rating of 3 as “general” ones. In this way, η is still calculated to be less than 1. This is because our current classification scheme (i.e. influencers with a rating of 3 are considered as “niche” ones), in fact, already overestimates η as it inflates the numerator and decreases the denominator.



Note: For each influencer, we define **generality_max** (**generality_min**) as the maximum (minimum) of the ratings on generality given by the two coders who reviewed this influencer. A larger circle indicates a higher number of observations.

Figure A1: Inter-coder Reliability on Generality Rating

clicking the influencer’s “store window”, one can see how many products she recommends. As described above, our data includes the following information: (i) whether the influencer has a “store window”, denoted as a dummy variable **product** whose value is 1 (0) if there is (not) a “store window” on the influencer’s homepage; (ii) how many products the influencer recommends, denoted as a numerical variable **products_recommend**.

To test the hypothesis that general influencers are more likely to participate in on-the-platform marketing campaigns, we regress **product** or **products_recommend** on an influencer’s **generality**. Since **product** is a dummy variable, a logit model is used for the regression with **product** as the dependent variable. Also, since the distribution of **products_recommend** in our dataset is extremely skewed (skewness = 6.94), we take log of (**products_recommend** + 1) as the dependent variable in the corresponding regression. Formally, we run the following regressions:

$$\log \left[\frac{\Pr(\text{product} = 1)}{\Pr(\text{product} = 0)} \right] = \beta_0 + \beta_1 \text{generality}, \quad (\text{A13})$$

and

$$\log(\text{products_recommend} + 1) = \beta_0 + \beta_1 \text{generality} + \varepsilon. \quad (\text{A14})$$

Our theoretical model predicts that the sign of β_1 , the coefficient before **generality**, is positive. As a robustness check, we can also include two control variables: (1) **num_followers**, which is the number of followers each influencer has, and (2) coder individual fixed effects. We summarize our results in Table A1.

Table A1: Regression Results

	<i>Dependent variable:</i>					
	product			log(products_recommend + 1)		
		<i>logistic</i>			<i>OLS</i>	
generality	0.633*** (0.176)	0.620*** (0.177)	0.672*** (0.181)	0.363*** (0.120)	0.378*** (0.121)	0.405*** (0.122)
num_followers		0.000 (0.000)	0.000 (0.000)		−0.000 (0.000)	−0.000 (0.000)
Constant	0.736*** (0.137)	0.634*** (0.202)	0.598** (0.256)	1.257*** (0.099)	1.373*** (0.135)	1.334*** (0.169)
Coder fixed effect	No	No	Yes	No	No	Yes
Observations	752	752	752	752	752	752
R ²				0.012	0.014	0.019
Adjusted R ²				0.011	0.011	0.013
Log Likelihood	−409.775	−409.533	−406.429			
Akaike Inf. Crit.	823.550	825.066	824.859			
<i>Note:</i>				*p<0.1; **p<0.05; ***p<0.01		

We see that the coefficient for **generality** is significantly positive. That is, general influencers are more likely to participate in on-the-platform marketing than niche ones, which is consistent with the prediction given by our theoretical model. Note that the result is robust to controlling for the number of followers and coder fixed effects. In fact, **num_followers** is not a statistically or economically significant predictor of our dependent variable (**product** or **products_recommend**) at all.

Note that there are alternative ways to construct the variable **generality**: classifying an influencer as “niche” if the average rating is *strictly* less than 3 (i.e., classifying influencers with an average rating 3 as “general”), or simply treating **generality** as a continuous variable ranging between 1 to 5. Table A2 shows that our result that the coefficient of **generality** is always positive and significant, regardless of how we construct the variable

generality.³¹

Table A2: Regression Results: Robustness Check

	<i>Dependent variable:</i>			
	product		log(products_recommend + 1)	
	<i>logistic</i>		<i>OLS</i>	
generality (rating < 3 as “niche”)	0.703*** (0.202)		0.419*** (0.145)	
generality (continuous)		0.299*** (0.098)		0.183*** (0.065)
Constant	0.588*** (0.176)	0.090 (0.351)	1.161*** (0.130)	0.847*** (0.240)
Observations	752	752	752	752
R ²			0.011	0.010
Adjusted R ²			0.010	0.009
Log Likelihood	−410.339	−411.514		
Akaike Inf. Crit.	824.678	827.027		
<i>Note:</i>			*p<0.1; **p<0.05; ***p<0.01	

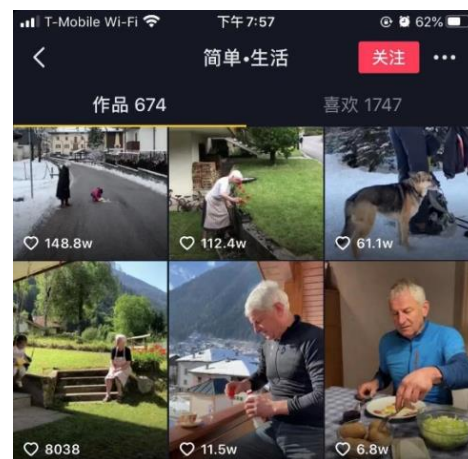
³¹We do not include control variables for this robust check since they have already been shown not to change the result.

Figure A2: Original Transcript of Instructions to Coders

1. 打开“抖音”
2. 点击右上角“搜索”
3. 输入 column B 中的抖音号 (“douyin_ID”), 比如“daijiali218”
4. 进入该主播主页 (进入前请 double check 主播的名字和 column C 是否一致)
5. 判断这是否是一个官方媒体 (比如“人民日报”、“XX 交通广播”等), 如果是, 在“official”一栏 (column D) 中填“1”, 否则填“0”。如果填了“1”, 请跳过 6-10 步。【注: 我们发给大家的主播中只有很少的是 official】
6. 在“抖音号: xxxxxx”下方请关注该主播是否有一栏名为“商品橱窗”的字样, 若答案为是, 请在“product”一栏 (column E) 中记录 1, 若答案为否, 请在此栏中记录 0。如果填了“0”, 请跳过第 7 步。
7. 请点击进入“商品橱窗”, 上面有两栏, 分别显示了“我的”和“推荐”各有几个产品。请在表格中的“Products_own” (column F) 和 “Products_recommend” (column G) 两栏里分别记录这两个数据。在下面截图的例子中, “Products_own”应为 0, “Products_recommend”应为 8。



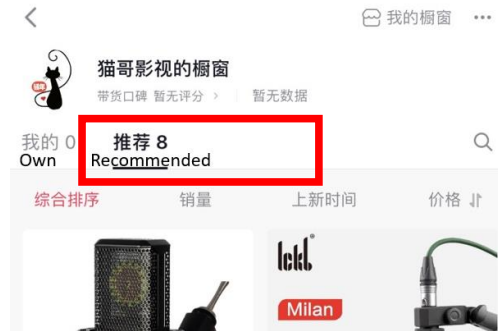
8. 点击左上角的箭头回到该主播的主页, 请大致浏览 (无需点进作品页面) 50 条该主播的近期视频作品【注意: 不是他商品橱窗里的商品!】, 并用 2 个词概括该主播的作品内容, 比如“小游戏”, “烹饪”, “美食”, “数码”等等。填入“type”一栏 (column H)
9. 基于你的判断, 请对该主播的“视频内容”广度按 1-5 打分, 填在“generality”一栏 (column I) 中:
 - a. 1 分是非常小众/单一 (例如某主播主页的作品全部是毛笔书法, 受众主要是书法爱好者, 下图左),
 - b. 5 分是非常大众/广泛 (例如某主播分享自己的衣食住行, 包含各种不同类别的内容, 受众可以是各种各样的人, 下图右)。



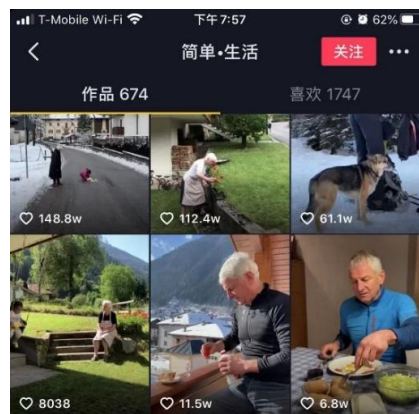
10. 如果这个主播是你熟知的影视或歌手明星 (比如迪丽热巴、陈赫等), 请在“celebrity”一栏 (column J) 中标注 1, 如果不是则不做操作。
11. 点击左上角的箭头回到上一页
12. 针对其他列表中的网红重复以上 2-11 步
13. 如果有任何问题, 请记录在 note (column K) 一栏中

Figure A3: Translated Transcript of Instructions to Coders

1. Open “Douyin.”
2. Click “search” on the top right corner.
3. Type in the douyin ID provided in column B, for example, “daijiali218.”
4. Enter the homepage of the influencer (Please double check whether the name of the influencer is the same as that in column C).
5. Decide whether this is an official channel (for example, “People’s Daily” or “XX City Traffic Radio”). If so, please fill “1” in the “official” column (column D), otherwise “0”. If you fill in “1”, please skip steps 6-10. [Note: There are only a few official channels.]
6. Under “Douyin ID: xxxxxxx”, please check if the influencer has her “store window.” If so, please record “1” in the “product” column (column E), otherwise “0”. If you have recorded “0” in this step, please skip step 7.
7. Please enter “store window,” where it says how many products the influencer “owns” and “recommends.” Please record these two numbers in “Products_own” (column F) and “Products_recommend” (column G), respectively. In the example below, “Products_own” is 0 and “Products_recommend” is 8.



8. Click the arrow in the top left corner and go back to the homepage of the influencer. Please scan and skim (no need to click into) 50 videos that the influencer uploaded recently [Note: We are talking about videos, not the products endorsed by the influencer]. Please summarize the content of the influencer’s videos in 1 or 2 words, for example, “gaming,” “cooking,” “food,” “digital products,” etc. Record this information in the “type” column (column H).
9. Based on your judgment, please rate, on a 1-5 scale, how general the content of this influencer’s videos is, and record your rating in the “generality” column (column I):
 - a. Score = 1 means very niche/concentrated (for example, some influencer only posts videos on calligraphy, so most of her followers would be fans of Chinese calligraphy: see the left picture below),
 - b. Score = 5 means very general/diverse (for example, some influencer posts videos about his life, and his followers can be very diverse: see the right picture below).



10. If this influencer is a celebrity (singer or actor) that you know, please record “1” in the column “celebrity” (column J).
11. Click the arrow on the top left corner to return to the previous page.
12. Repeat steps 2-11 for each influencer on your list.
13. If there are any issues about an influencer, please take a note in column K.